Proper Title

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Abstract

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Keywords: Keyords go here.
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1 Introduction

1.1 Existing literature

The literature on singularities in finite and nonlinear hyperelasticity is organised around three major phenomena.

- **Crack tips in tension** (Knowles and Sternberg, 1970s). The classical work on nonlinear singularities treats a 360° corner (a crack) pulled in pure tension. In a neo-Hookean material the asymptotic displacement field departs markedly from linear fracture mechanics; the analysis is, however, confined to traction-free boundaries and pure tensile opening.

- **Cavitation** (Ball, 1982). Under severe hydrostatic tension the hyperelastic equations admit singular weak solutions in which a hole opens spontaneously in the interior of a solid domain — a point singularity with $\boldsymbol{F} \to \infty$.

- **Surface creasing** (Biot, 1963; Hong et al., 2009). A compressed flat half-space of neo-Hookean material becomes unstable at a critical stretch ($\lambda \approx 0.54$); this instability is now understood as a *crease*, a point singularity nucleating on an initially flat surface.

1.2 The gap

The existing nonlinear-singularity literature largely avoids the configuration of interest here: *a geometric corner (e.g.* 90°*) carrying mixed boundary conditions (Dirichlet meeting Neumann) under compression.* Each strand addresses only part of it:

- Linear theory (Kondratiev) handles the mixed-BC corner but ignores the large-deformation kinematics.

- Nonlinear theory (Knowles–Sternberg) handles large deformations but avoids the Dirichlet–Neumann condition and avoids compression.

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- Instability theory handles compression but typically on flat domains or symmetric geometries, ignoring a pre-existing geometric corner singularity.

The open question this paper addresses is how an already-present stress concentration at a geometric Dirichlet–Neumann corner couples with a nonlinear structural instability (buckling/creasing), and how the resulting tangent operators can be bounded.

2 Problem formulation

2.1 Finite-strain elastostatics as energy minimisation

We consider a hyperelastic body occupying a reference d-dimensional domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, $\Gamma_D \cap \Gamma_N = \emptyset$. The deformation of the body is described by the displacement field $\boldsymbol{u} : \Omega \to \mathbb{R}^d$, the deformation map $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{x} + \boldsymbol{u}(\boldsymbol{x})$, and the associated deformation gradient

$$\boldsymbol{F} = \nabla \boldsymbol{y} = \boldsymbol{I} + \nabla \boldsymbol{u}, \qquad J = \det \boldsymbol{F} > 0, \tag{1}$$

where ∇ denotes the gradient with respect to the reference configuration. The mechanical response is encoded in a strain energy density $\Psi(\boldsymbol{F})$, and the equilibrium configuration is the minimiser of the total potential energy

$$\mathcal{E}(\boldsymbol{u}) = \int_\Omega \Psi(\boldsymbol{F})\, dV - \int_\Omega \boldsymbol{f}_0 \cdot \boldsymbol{u}\, dV - \int_{\Gamma_N} \boldsymbol{t}_0 \cdot \boldsymbol{u}\, dS, \tag{2}$$

over the space of admissible displacements $U = \{\boldsymbol{u} \in H^1(\Omega; \mathbb{R}^d) : \boldsymbol{u} = \boldsymbol{g} \text{ on } \Gamma_D\}$, where $\boldsymbol{f}_0$ is a body force, $\boldsymbol{t}_0$ a prescribed traction on the Neumann boundary, and $\boldsymbol{g}$ the prescribed boundary displacement. We seek $\boldsymbol{u} \in U$ such that $\mathcal{E}(\boldsymbol{u}) \le \mathcal{E}(\boldsymbol{v})$ for all admissible $\boldsymbol{v}$.

2.2 Stationarity, stress, and tangent modulus

A minimiser satisfies the stationarity condition $D_{\delta\boldsymbol{u}}\mathcal{E}(\boldsymbol{u}) = 0$ for all admissible variations $\delta\boldsymbol{u}$ (vanishing on Γ_D), which is the weak form of equilibrium

$$\int_\Omega \boldsymbol{P}(\boldsymbol{F}) : \nabla\delta\boldsymbol{u}\, dV = \int_\Omega \boldsymbol{f}_0 \cdot \delta\boldsymbol{u}\, dV + \int_{\Gamma_N} \boldsymbol{t}_0 \cdot \delta\boldsymbol{u}\, dS, \tag{3}$$

where the first Piola–Kirchhoff stress is the first derivative of the energy with respect to the deformation gradient,

$$\boldsymbol{P}(\boldsymbol{F}) = \frac{\partial \Psi}{\partial \boldsymbol{F}}. \tag{4}$$

The natural boundary condition associated with Γ_N is $\boldsymbol{P}\boldsymbol{N} = \boldsymbol{t}_0$, with $\boldsymbol{N}$ the reference outward unit normal. Because (3) is nonlinear in $\boldsymbol{u}$, it is solved iteratively by Newton's method. Linearising the residual about a current iterate $\bar{\boldsymbol{u}}$ in a direction $\Delta\boldsymbol{u}$ introduces the fourth-order tangent (elasticity) modulus $\mathbb{L}$, with components L_{iJkL},

$$\mathbb{L} = \frac{\partial \boldsymbol{P}}{\partial \boldsymbol{F}} = \frac{\partial^2 \Psi}{\partial \boldsymbol{F} \otimes \partial \boldsymbol{F}}, \qquad L_{iJkL} = \frac{\partial^2 \Psi}{\partial F_{iJ}\, \partial F_{kL}}, \tag{5}$$

which possesses the major symmetry $L_{iJkL} = L_{kLiJ}$, so that each Newton step requires solving the linear problem

$$\int_\Omega \nabla\delta\boldsymbol{u} : \mathbb{L}(\bar{\boldsymbol{F}}) : \nabla\Delta\boldsymbol{u}\, dV = -\, R(\bar{\boldsymbol{u}}; \delta\boldsymbol{u}), \tag{6}$$

with $R(\bar{\boldsymbol{u}}; \delta\boldsymbol{u})$ the residual of (3) evaluated at $\bar{\boldsymbol{u}}$. The well-posedness and conditioning of (6)—and hence the convergence of the whole nonlinear solver—are governed by the coercivity and boundedness of the bilinear form induced by $\mathbb{L}(\bar{\boldsymbol{F}})$. The central object of this work is therefore the behaviour of $\mathbb{L}$ as a function of the local deformation state.

2.3 The neo-Hookean material

To fix ideas and to exhibit explicit constants we use, as a running example, the compressible neo-Hookean strain energy density adopted in the cut-finite-element study [9] that motivates this analysis,

$$\Psi(\boldsymbol{F}) = \frac{\mu_e}{2}\big(\mathrm{tr}(\boldsymbol{F}^T\boldsymbol{F}) - d\big) + f(J), \tag{7}$$

where $\mu_e > 0$ is the shear modulus and $f(J)$ is a strictly convex volumetric penalty with $f(1) = 0$, $f'(1) = -\mu_e$, and $f(J) \to \infty$ as $J \to 0^+$. Two standard choices are

$$f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J, \qquad f_{\mathrm{quad}}(J) = \tfrac{\kappa}{2}(J-1)^2, \tag{8}$$

with λ_e the Lamé constant and κ a bulk modulus; (7) with $f_{\log}$ reproduces the model of [9]. The corresponding stress and tangent are

$$\boldsymbol{P}(\boldsymbol{F}) = \mu_e\boldsymbol{F} + f'(J)\,J\,\boldsymbol{F}^{-T}, \tag{9}$$

$$L_{iJkL} = \mu_e\,\delta_{ik}\delta_{JL} + \big(f''(J)J^2 + f'(J)J\big)\,F^{-T}_{iJ}F^{-T}_{kL} - f'(J)J\,F^{-T}_{iL}F^{-T}_{kJ}. \tag{10}$$

The first term $\mu_e\,\delta_{ik}\delta_{JL}$ is constant; the entire nonlinear part is controlled by J and the *inverse* deformation gradient $\boldsymbol{F}^{-1}$, yielding the structural bound

$$\|\mathbb{L}(\boldsymbol{F})\| \le \mu_e + \Theta(J)\,\|\boldsymbol{F}^{-1}\|^2, \tag{11}$$

where $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1}$ is set by the smallest principal stretch and Θ is a scalar growth function fixed by f (computed explicitly in §3.4). Consequently $\mathbb{L}$ can become unbounded through volumetric collapse, $\lambda_{\min} \to 0$ (equivalently $\boldsymbol{F}^{-1} \to \infty$); whether a purely tensile blow-up of $\boldsymbol{F}$ (with $\boldsymbol{F}^{-1}$ bounded) leaves $\mathbb{L}$ finite depends on the growth of $\Theta(J)$, which turns out to discriminate sharply between the two penalties in (8). This dichotomy is the analytical backbone of the present paper.

2.4 The mixed-corner singularity

The configuration of interest is a right-angle junction where the two boundary-condition types meet: a Dirichlet face (Γ_D, prescribed displacement) abutting a traction-free Neumann face (Γ_N). In $\mathbb{R}^d$ this junction is a codimension-two *edge* Σ (a point for $d = 2$, a curve or surface for $d \ge 3$); its transverse geometry, in the two-plane normal to Σ, is the right-angle corner we analyse, and we use polar coordinates (r, θ) in that transverse plane. Such junctions arise generically wherever a clamped face meets a traction-free face — for instance along the base edges of a compressed body, the concrete setting taken up in Section 5. Classical Kondratiev/edge theory for the linearised problem predicts a transverse singularity

$$\boldsymbol{u}(r,\theta) \sim K\,r^{\alpha}\,\boldsymbol{\Phi}(\theta), \qquad \alpha \in (0,1), \tag{12}$$

with r the distance to the edge Σ, K a generalised stress-intensity factor and α the (geometry- and Poisson-ratio-dependent) singularity exponent. The deformation gradient then scales as $\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}(Kr^{\alpha-1})$, so $\|\boldsymbol{F}\| \to \infty$ as $r \to 0$.

The question this raises—and which the remainder of the paper addresses—is how this kinematic singularity propagates into the nonlinear tangent operator (5). In view of (11), the answer hinges on which principal stretch diverges: a tensile corner keeps $\boldsymbol{F}^{-1}$ bounded and (for a suitable penalty) leaves $\mathbb{L}$ well behaved, whereas a compressive corner drives $\lambda_{\min} \to 0$ and blows up the *norm* of $\mathbb{L}$ (severe ill-conditioning). The tangent nonetheless remains strongly elliptic — the compressive instability is *structural* (a corner analogue of creasing), not a change of type of the PDE. The following sections analyse the two scenarios in turn (Section 3) and then show how a small imperfection regularises the compressive case through a structural bifurcation (Section 4).

3 Asymptotic behaviour of the tangent operator at a mixed corner

3.1 Geometry and linearised kinematics

Let the reference domain $\Omega \subset \mathbb{R}^d$ ($d \geq 2$) carry a right-angle Dirichlet–Neumann edge Σ of codimension two. Near a point of Σ split $\mathbb{R}^d = T\Sigma \oplus \Pi$ into the edge tangent space and the normal two-plane Π, and use polar coordinates (r, θ) in Π with $r \in [0, R]$, $\theta \in [0, \pi/2]$, where $r = \operatorname{dist}(\cdot, \Sigma)$. Write B_ρ for the tubular neighbourhood $\{\boldsymbol{x} \in \Omega : \operatorname{dist}(\boldsymbol{x}, \Sigma) < \rho\}$ of the edge. The leading singularity is governed by the transverse problem in Π, modulated smoothly along Σ; for $d = 2$ the edge degenerates to a point and B_ρ to the corner disc.

- **Dirichlet boundary (Γ_D):** $\theta = 0$, deformation prescribed, $\boldsymbol{y}(r, 0) = \boldsymbol{g}(r)$.
- **Neumann boundary (Γ_N):** $\theta = \pi/2$, traction-free, $\boldsymbol{P}\boldsymbol{N} = \boldsymbol{0}$.

Classical Kondratiev theory for the *linearised* problem gives the leading-order singular displacement

$$\boldsymbol{u}(r, \theta) \sim K\, r^{\alpha}\, \boldsymbol{\Phi}(\theta), \tag{13}$$

with K the generalised stress-intensity factor (set by the far field), $\boldsymbol{\Phi}$ a smooth angular profile, and $\alpha \in (0, 1)$ the geometry- and Poisson-dependent exponent. Consequently

$$\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}\big(K\, r^{\alpha - 1}\big), \qquad \|\boldsymbol{F}\| \to \infty \text{ as } r \to 0. \tag{14}$$

Whether this kinematic blow-up contaminates the tangent operator $\mathbb{L} = \partial^2 \Psi / \partial \boldsymbol{F}\, \partial \boldsymbol{F}$ depends not on $\|\boldsymbol{F}\|$ but on which principal stretches diverge, which is the content of the rest of this section.

3.2 Standing assumptions on the stored energy

The mechanism we describe is not specific to the neo-Hookean model; that model only serves to fix ideas and to exhibit explicit constants. We therefore phrase the analysis for a general frame-indifferent energy and isolate the structural properties that the argument actually uses. Throughout, $\|\cdot\|$ denotes the operator (spectral) norm on tensors, $0 < \lambda_{\min}(\boldsymbol{F}) \leq \cdots \leq \lambda_{\max}(\boldsymbol{F})$ are the singular values of $\boldsymbol{F}$ (the d principal stretches), and we recall $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}(\boldsymbol{F})^{-1}$.

Assumption 1 (Regularity and frame indifference)**.** The stored energy $\Psi : \mathrm{GL}^+(d) \to \mathbb{R}$ is objective, $\Psi \in C^2(\mathrm{GL}^+(d))$, and $\Psi(\boldsymbol{F}) \to +\infty$ as $\det \boldsymbol{F} \to 0^+$.

Assumption 2 (Inverse-channel growth of the tangent)**.** There exist a continuous, nondecreasing function $\Theta : (0, \infty) \to (0, \infty)$ and a constant $c_0 \geq 0$ such that the tangent modulus $\mathbb{L}(\boldsymbol{F}) = D^2\Psi(\boldsymbol{F})$ satisfies, for all $\boldsymbol{F} \in \mathrm{GL}^+(d)$,

$$\|\mathbb{L}(\boldsymbol{F})\| \;\leq\; c_0 \;+\; \Theta\big(\det \boldsymbol{F}\big)\, \|\boldsymbol{F}^{-1}\|^2. \tag{15}$$

Assumption 3 (Uniform strong ellipticity on bounded-distortion sets)**.** For every $0 < c \le 1$ and $M \ge 1$ there is $\gamma(c, M) > 0$ such that whenever $\lambda_{\min}(\boldsymbol{F}) \ge c$ and $\det \boldsymbol{F} \le M$,

$$L_{iJkL}(\boldsymbol{F})\, m_i N_J\, m_k N_L \;\ge\; \gamma(c, M)\, |\boldsymbol{m}|^2\, |\boldsymbol{N}|^2 \qquad \forall\, \boldsymbol{m}, \boldsymbol{N} \in \mathbb{R}^2. \tag{16}$$

The decisive feature of Assumption 2 is that the inverse-stretch factor $\|\boldsymbol{F}^{-1}\|^2 = \lambda_{\min}^{-2}$ is the *only* algebraic channel through which the Hessian can blow up; the dependence on $\|\boldsymbol{F}\|$ (equivalently on $\lambda_{\max}$) enters solely through the scalar $\Theta(\det \boldsymbol{F})$. Whether a tensile corner ($\lambda_{\max} \to \infty$, $\det \boldsymbol{F} \to \infty$, $\lambda_{\min}$ bounded below) is benign therefore reduces to the *growth rate* of Θ, which we quantify in §3.4. Materials with genuine tensile stiffening of the deviatoric part (Gent, Arruda–Boyce), whose tangent diverges through $\lambda_{\max}$ directly rather than through $\Theta(\det \boldsymbol{F})$, violate (15) and are excluded.

Lemma 1 (Assumptions for the neo-Hookean model)**.** *Let* Ψ *be the neo-Hookean energy* (7) *with* $\mu_e > 0$ *and* $f \in C^2(0, \infty)$*. Then Assumption 2 holds with* $c_0 = \mu_e$ *and*

$$\Theta(J) = \big|\, f''(J) J^2 + f'(J) J \,\big| + \big|\, f'(J) J \,\big|, \tag{17}$$

and Assumption 3 holds (rank-one convexity away from $\det \boldsymbol{F} = 0$*). Assumption 1 additionally requires the barrier property* $f(J) \to \infty$ *as* $J \to 0^+$*: this holds for* $f_{\log}$ *but* fails *for* f_{quad}*, since* $f_{\text{quad}}(J) \to \frac{\kappa}{2} < \infty$ *(see Remark 4 and Scenario B).*

Proof. Objectivity and C^2 regularity on $\mathrm{GL}^+(d)$ are immediate; the barrier hypothesis $f(J) \to \infty$ as $J \to 0^+$, when assumed, gives the blow-up required by Assumption 1. In (10) the first term is the scaled identity, of norm μ_e. Each of the remaining two terms is a rank-one, respectively transposition, contraction of $\boldsymbol{F}^{-T} \otimes \boldsymbol{F}^{-T}$; since $\|\boldsymbol{F}^{-T}\| = \|\boldsymbol{F}^{-1}\|$, each is bounded in operator norm by its scalar prefactor times $\|\boldsymbol{F}^{-1}\|^2$. The triangle inequality gives (15) with $c_0 = \mu_e$ and Θ as in (17). For Assumption 3, contracting (10) with $\boldsymbol{m} \otimes \boldsymbol{N}$ and writing $\xi := \boldsymbol{m} \cdot \boldsymbol{F}^{-T} \boldsymbol{N} = \boldsymbol{N} \cdot \boldsymbol{F}^{-1} \boldsymbol{m}$, the two volumetric terms combine $((f'' J^2 + f' J)\xi^2 - f' J\, \xi^2)$ to the single clean form

$$L_{iJkL} m_i N_J m_k N_L = \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 + f''(J)\, J^2\, \xi^2. \tag{18}$$

Wherever $f''(J) \ge 0$ the second term is nonnegative and (16) holds with $\gamma = \mu_e$, uniformly in $\boldsymbol{F}$ (no distortion bound needed); this covers f_{quad} ($f'' = \kappa > 0$) everywhere and $f_{\log}$ on $\{f''_{\log} \ge 0\}$. □ □

Remark 1 (Strength and a tension caveat). Identity (18) is sharper than (16): the neo-Hookean tangent is strongly elliptic with constant *exactly* μ_e wherever $f''(J) \ge 0$, and the volumetric term only adds to it. The Hessian of the quadratic (Hookean) part is constant, which is why the *deviatoric* response never blows up; only the *volumetric* response, through Θ, can. One caveat for $f_{\log}$: since $f''_{\log}(J) = (\mu_e + 2\lambda_e - 2\lambda_e \log J)/J^2$, convexity is lost once $J > J_* := \exp\big(1 + \mu_e/(2\lambda_e)\big)$, and then (18) can turn negative for ξ near its maximal value $|\boldsymbol{m}||\boldsymbol{N}|/\lambda_{\min}$. Hence Assumption 3 holds throughout *compression* ($J \le 1$) and moderate tension, but $f_{\log}$ can lose strong ellipticity under *extreme dilation* ($J > J_*$)—a tension-side phenomenon distinct from the compressive analysis, and one more reason the bounded-J qualifier matters in §3.5. The globally convex f_{quad} is free of this.

3.3 The non-compressive hypothesis

The qualitative split between the two scenarios is captured by a single quantity: the smallest principal stretch near the corner. We call the corner singularity *non-compressive* when $\lambda_{\min}$ stays bounded away from zero, even though $\lambda_{\max}$ (hence $\|\boldsymbol{F}\|$ and $\det \boldsymbol{F}$) may diverge.

Assumption 4 (Non-compressive singularity, (NC))**.** There exist $\rho \in (0, R]$ and $c \in (0, 1]$ such that the equilibrium deformation satisfies

$$\lambda_{\min}(\boldsymbol{F}(\boldsymbol{x})) \geq c, \quad \text{equivalently} \quad \|\boldsymbol{F}^{-1}(\boldsymbol{x})\| \leq c^{-1}, \qquad \text{for a.e. } \boldsymbol{x} \in B_\rho. \tag{19}$$

We stress what (19) does *not* say. It does not require incompressibility, and crucially it does *not* bound $\det \boldsymbol{F}$: in a tensile corner $\lambda_{\max} \sim r^{\alpha-1} \to \infty$, so $J = \det \boldsymbol{F} = \prod_i \lambda_i \sim r^{\alpha-1} \to \infty$ as well (one diverging stretch, the remaining $d-1$ bounded). The only quantity (NC) controls is $\|\boldsymbol{F}^{-1}\| \leq c^{-1}$. Physically, (NC) states that the material is pulled around the corner (tension/shear) rather than crushed into it. Because $\det \boldsymbol{F}$ is *not* bounded under (NC), the factor $\Theta(\det \boldsymbol{F})$ in (15) need not stay bounded; the next subsection makes its growth explicit, and this is precisely where the two volumetric penalties (8) part ways. Scenario B (§3.6) treats the complementary, compressive case in which (19) fails.

3.4 Explicit volumetric growth functions

The bound (15) exposes *two independent channels* through which the tangent can diverge, and they are kinematically decoupled:

$$\|\mathbb{L}(\boldsymbol{F})\| \leq \mu_e + \underbrace{\Theta(\det \boldsymbol{F})}_{\substack{\text{dilation channel} \\ J\to\infty}} \cdot \underbrace{\|\boldsymbol{F}^{-1}\|^2}_{\substack{\text{inverse channel} \\ \lambda_{\min}\to 0}}. \tag{20}$$

The *inverse channel* is the one usually blamed for trouble: it requires $\lambda_{\min} \to 0$, i.e. compression to zero volume, and is the subject of Scenario B. The *dilation channel*, $\Theta(\det \boldsymbol{F}) \to \infty$ as $J \to \infty$, is activated by *expansion* and is precisely what a tensile corner switches on, since $\lambda_{\max} \sim r^{\alpha-1} \to \infty$ forces $J \to \infty$ even with $\lambda_{\min}$ bounded below by (NC). Whether this matters is therefore not a property of the corner or of the sign of the loading, but of the *growth rate of* Θ—a property of the volumetric penalty f alone. We now evaluate it for the two penalties (8); the contrast shows that "tension is benign" is a model-dependent statement.

Logarithmic penalty. For $f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J$,

$$f'(J)J = -\mu_e + 2\lambda_e \log J, \qquad f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J, \tag{21}$$

so that the combination collapses to a constant,

$$f''(J)J^2 + f'(J)J = 2\lambda_e, \tag{22}$$

and therefore

$$\Theta_{\log}(J) = 2\lambda_e + \big|\, 2\lambda_e \log J - \mu_e \,\big| = \mathcal{O}\big(\log J\big) \quad \text{as } J \to \infty. \tag{23}$$

The volumetric stiffening grows only *logarithmically* in J.

Quadratic penalty. For $f_{\text{quad}}(J) = \frac{\kappa}{2}(J-1)^2$,

$$f'(J)J = \kappa J(J-1), \qquad f''(J)J^2 = \kappa J^2, \qquad f''(J)J^2 + f'(J)J = \kappa J(2J-1), \tag{24}$$

so

$$\Theta_{\text{quad}}(J) = \kappa J\,|2J-1| + \kappa J\,|J-1| = \mathcal{O}\big(\kappa J^2\big) \quad \text{as } J \to \infty. \tag{25}$$

The volumetric stiffening grows *quadratically* in J.

Consequence at the corner. We combine (15) with (NC) ($\|\boldsymbol{F}^{-1}\|^2 \le c^{-2}$) and the tensile kinematics. In the non-compressive regime exactly one stretch diverges: $\lambda_{\max} \sim |K|\, r^{\alpha-1}$ while the remaining $d-1$ stretches stay bounded (in particular $\lambda_{\min}$ bounded below), so $J = \det \boldsymbol{F} = \prod_i \lambda_i \sim r^{\alpha-1}$. (If instead $\nabla \boldsymbol{u}$ had two diverging eigenvalues, the corresponding subdeterminant would give $J \sim r^{2(\alpha-1)}$; we work in the one-large-stretch regime characteristic of a Dirichlet–Neumann opening, where $J \sim r^{\alpha-1}$.) The tangent norm near the corner then obeys

$$\|\mathbb{L}(\boldsymbol{x})\| \le \mu_e + c^{-2}\,\Theta\big(\det \boldsymbol{F}(\boldsymbol{x})\big) \sim \begin{cases} \mu_e + \mathcal{O}\big(\log \frac{1}{r}\big), & f = f_{\log}, \\ \mathcal{O}\big(r^{-2(1-\alpha)}\big), & f = f_{\text{quad}}. \end{cases} \tag{26}$$

Thus the logarithmic penalty keeps the tangent bounded up to a logarithm, whereas the quadratic penalty produces a genuine *algebraic* blow-up of the tangent *even in pure tension*. This is the dilation channel of (20) acting alone, with the inverse channel inert ($\|\boldsymbol{F}^{-1}\|$ bounded); it is algebraically singular of a severity comparable to—though mechanistically distinct from—the compressive blow-up of Scenario B, which instead runs through $\|\boldsymbol{F}^{-1}\|^2$. The often-repeated statement that "tension is harmless for neo-Hookean" is therefore an artefact of the logarithmic volumetric model and is false for the quadratic one.

Physical reading: the large-dilation response. The dichotomy is, at bottom, a statement about how the bulk energy behaves under unbounded *expansion*, the loading a tensile corner imposes:

volumetric model	energy as $J \to \infty$	tangent $\Theta(J)$
$\lambda_e \log^2 J$	$\sim \log^2 J$ (soft)	$\mathcal{O}(\log J)$
$\frac{\kappa}{2}(J-1)^2$	$\sim J^2$ (stiff)	$\mathcal{O}(J^2)$

A penalty that is polynomial in J makes the material resist further dilation ever more stiffly, and the linearised modulus faithfully reports that stiffness as a blow-up; the logarithmic penalty is gentle about expansion and does not. Both penalties are designed to diverge correctly as $J \to 0$ (resisting collapse, the inverse channel); their large-J behaviour, which the dilation channel probes, was never physically constrained, and that latitude is exactly where they part. The exact cancellation (22) is the analytic signature of this softness. We make the benign case precise in Theorem 1 and flag the algebraic case in Remark 4.

3.5 Scenario A: the tensile (non-compressive) singularity

We work under the mild-growth condition isolated above.

Assumption 5 (Mild volumetric growth)**.** The growth function satisfies $\Theta(J) = \mathcal{O}\big((\log J)^p\big)$ as $J \to \infty$ for some $p \ge 0$.

By (23), $f_{\log}$ satisfies Assumption 5 with $p = 1$; by (25), f_{quad} does not.

Lemma 2 (Subordinate tangent)**.** *Under Assumptions 1, 2, 5 and (NC), for every $\varepsilon > 0$ there is C_ε with*

$$\|\mathbb{L}(\boldsymbol{x})\| \le C_\varepsilon\, r^{-\varepsilon} \qquad \text{for a.e. } \boldsymbol{x} \in B_\rho. \tag{27}$$

In particular $\mathbb{L} \in L^q(B_\rho)$ for every $q < \infty$, and $\|\mathbb{L}\|$ is dominated by any negative power of r.

Proof. By (15) and (NC), $\|\mathbb{L}\| \le \mu_e + c^{-2}\Theta(\det \boldsymbol{F})$. Assumption 5 and $\det \boldsymbol{F} = \mathcal{O}(r^{\alpha-1})$ give $\Theta(\det \boldsymbol{F}) = \mathcal{O}\big((\log \frac{1}{r})^p\big)$, and $(\log \frac{1}{r})^p \le C_\varepsilon r^{-\varepsilon}$ on $(0, \rho]$ for every $\varepsilon > 0$. □ □

Lemma 3 (Strong ellipticity outside the dilation core)**.** *Let $f = f_{\log}$ and let $J_* = \exp\big(1+\mu_e/(2\lambda_e)\big)$ be the dilation threshold of Remark 1, beyond which $f''_{\log} < 0$. Under Assumptions 1, 3 and (NC), with the tensile kinematics $J \sim |K|\, r^{\alpha-1}$ of §3.4, define the* dilation core radius

$$r_* := \big(J_*/|K|\big)^{1/(\alpha-1)} = \big(|K|/J_*\big)^{1/(1-\alpha)}, \tag{28}$$

so that $J(\boldsymbol{x}) \le J_*$ for $r \ge r_*$. Then on the annular region $A_* := \{\boldsymbol{x} \in B_\rho : r_* < r < \rho\}$ the tangent satisfies the Legendre–Hadamard condition with the constant *ellipticity modulus μ_e,*

$$L_{iJkL}(\boldsymbol{x})\, m_i N_J m_k N_L \;\ge\; \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 \qquad \textit{for a.e. } \boldsymbol{x} \in A_*, \quad \forall \boldsymbol{m}, \boldsymbol{N} \in \mathbb{R}^2. \tag{29}$$

On the complementary core $\{r \le r_*\}$, where $J > J_*$, strong ellipticity of $f_{\log}$ is genuinely lost (Remark 1): the bound (29) *fails and the modulus $\mu_e + f''(J) J^2 \xi^2/(|\boldsymbol{m}|^2|\boldsymbol{N}|^2)$ can turn negative.*

Proof. By the exact rank-one identity (18),

$$L_{iJkL} m_i N_J m_k N_L = \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 + f''(J)\, J^2\, \xi^2, \qquad \xi := \boldsymbol{m} \cdot \boldsymbol{F}^{-T} \boldsymbol{N},$$

in which the volumetric prefactor is the single scalar $f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J$ (cf. (22)). This is nonnegative precisely when $\log J \le 1 + \mu_e/(2\lambda_e)$, i.e. $J \le J_*$. Since $J \sim |K| r^{\alpha-1}$ is decreasing in r (recall $\alpha < 1$), the threshold $J = J_*$ is crossed at the single radius (28), and $J \le J_*$ throughout A_*. There the second term only adds to μ_e, giving (29). For $r < r_*$ we have $J > J_*$, hence $f''(J)J^2 < 0$; taking $\boldsymbol{m}, \boldsymbol{N}$ aligned so that ξ^2 attains its maximum $|\boldsymbol{m}|^2 |\boldsymbol{N}|^2\, \|\boldsymbol{F}^{-1}\|^2$ (admissible since $\|\boldsymbol{F}^{-1}\| \le c^{-1}$ is only an upper bound, with equality possible) drives the form below zero once $|f''(J)J^2|\, \|\boldsymbol{F}^{-1}\|^2 > \mu_e$. □ □

The picture is therefore sharper, and more honest, than "tension is harmless": for the logarithmic penalty the tensile corner is strongly elliptic on the annulus A_* outside a *dilation core* $\{r \le r_*\}$, inside which $f_{\log}$ itself — not the corner — ceases to be strongly elliptic because the material is dilated past J_*. The core radius r_* in (28) shrinks as the threshold $J_* = \exp(1 + \mu_e/(2\lambda_e))$ grows, i.e. as the penalty is made stiffer against extreme dilation (large λ_e/μ_e); a model with $f'' \ge 0$ globally would have $r_* = 0$ and no core. This is the tensile mirror of the model-dependence already seen for Θ in §3.4, and it confines the genuinely delicate behaviour to a controlled inner region.

For the weighted well-posedness we recall two standard inequalities in the Kondratiev space $V^{1,2}_\beta(B_\rho)$—the closure of smooth functions vanishing on $\Gamma_D \cap \partial B_\rho$ under $\|\boldsymbol{v}\|^2_{V^{1,2}_\beta} = \int_{B_\rho} \big(r^{2\beta}|\nabla \boldsymbol{v}|^2 + r^{2\beta-2}|\boldsymbol{v}|^2\big)\, dV$.

Lemma 4 (Weighted Hardy and Korn inequalities [6, 8])**.** *Let $\beta \in \mathbb{R}$ avoid the (discrete) set of exceptional exponents of the corner. There are constants C_H, C_K depending only on β and the opening angle such that, for all $\boldsymbol{v} \in V^{1,2}_\beta(B_\rho)$ vanishing on $\Gamma_D \cap \partial B_\rho$,*

$$\int_{B_\rho} r^{2\beta-2} |\boldsymbol{v}|^2\, dV \le C_H \int_{B_\rho} r^{2\beta} |\nabla \boldsymbol{v}|^2\, dV, \tag{30}$$

$$\int_{B_\rho} r^{2\beta} |\nabla \boldsymbol{v}|^2\, dV \le C_K \int_{B_\rho} r^{2\beta}\, |\boldsymbol{\varepsilon}(\boldsymbol{v})|^2\, dV, \qquad \boldsymbol{\varepsilon}(\boldsymbol{v}) = \tfrac{1}{2}\big(\nabla \boldsymbol{v} + \nabla \boldsymbol{v}^T\big). \tag{31}$$

Let $\beta_\star$ be the weight exponent for which the *linear* elasticity corner problem at this Dirichlet–Neumann junction is Fredholm (equivalently $\beta_\star$ avoids the singular exponents α). The existence of such a weight, and the resulting Fredholm property in the weighted space $V^{1,2}_{\beta_\star}$, is the standard Kondratiev–Maz'ya–Nazarov–Plamenevsky theory for elliptic systems with conical/edge singularities [6, 8]: the operator is Fredholm precisely when $\beta_\star$ avoids the discrete spectrum of the corner operator pencil (the singular exponents), and the resolvent is then compact, so the spectrum is discrete and the critical mode $\boldsymbol{\phi}$ of Assumption 6 is genuinely attained. We may write the Newton-step bilinear form as

$$a(\Delta\boldsymbol{u},\delta\boldsymbol{u}) := \int_{B_\rho} \nabla\delta\boldsymbol{u} : \mathbb{L}(\bar{\boldsymbol{F}}) : \nabla\Delta\boldsymbol{u}\, dV. \tag{32}$$

Theorem 1 (Well-posedness of the linearised step, tensile corner)**.** *Let $f = f_{\log}$ and work on the annular corner neighbourhood $A_* = \{r_* < r < \rho\}$ outside the dilation core* (28) *of Lemma 3 (for a globally convex-volumetric barrier the core is empty, $r_* = 0$, and $A_* = B_\rho$). Under Assumptions 1–5 and the non-compressive hypothesis (NC), the form* (32)*, restricted to A_*, is bounded and coercive on $V^{1,2}_{\beta_\star}(A_*)$ modulo a compact perturbation: there exist constants $\Lambda < \infty$, $\gamma_1 > 0$ and a compact form $k(\cdot,\cdot)$ such that, for all $\boldsymbol{u},\boldsymbol{v} \in V^{1,2}_{\beta_\star}(A_*)$,*

$$|a(\boldsymbol{u},\boldsymbol{v})| \le \Lambda\, \|\boldsymbol{u}\|_{V^{1,2}_{\beta_\star}}\, \|\boldsymbol{v}\|_{V^{1,2}_{\beta_\star}}, \tag{33}$$

$$a(\boldsymbol{v},\boldsymbol{v}) \ge \gamma_1\, \|\boldsymbol{v}\|^2_{V^{1,2}_{\beta_\star}} - k(\boldsymbol{v},\boldsymbol{v}). \tag{34}$$

Consequently, on A_*, the linearised corner operator is Fredholm of index zero in $V^{1,2}_{\beta_\star}$; for $\beta_\star$ avoiding the corner spectrum it is an isomorphism, its solutions admit the asymptotic expansion (13)*, and the governing singular exponents coincide with those of the linearised elasticity problem. The inner core $\{r \le r_*\}$, where $f_{\log}$ loses strong ellipticity (Lemma 3), is a fixed-size region carrying no vertex; it must be treated separately and is the price of the logarithmic penalty in extreme tension (see Remark 2).*

Proof. Boundedness. By Cauchy–Schwarz and Lemma 2, on A_* the coefficient obeys the sharp bound $\|\mathbb{L}\| \le \mu_e + c^{-2}\Theta(J) = \mu_e + \mathcal{O}\big(\log\frac{1}{r}\big)$, a *logarithm*, not a power. Hence for every $\delta > 0$ there is C_δ with $\|\mathbb{L}\| \le C_\delta\, r^{-\delta}$, and

$$|a(\boldsymbol{u},\boldsymbol{v})| \le \int_{A_*} \|\mathbb{L}\|\, |\nabla\boldsymbol{u}|\, |\nabla\boldsymbol{v}|\, dV \le C_\delta \Big(\int_{A_*} r^{2\beta_\star} |\nabla\boldsymbol{u}|^2\Big)^{1/2} \Big(\int_{A_*} r^{-2\delta-2\beta_\star} |\nabla\boldsymbol{v}|^2\Big)^{1/2}.$$

This exhibits a as bounded from $V^{1,2}_{\beta_\star} \times V^{1,2}_{\beta_\star-\delta}$. The logarithm is subordinate to *every* algebraic weight, so the loss may be made smaller than the slack to the weight interval: let (β_-,β_+) be the maximal open interval of weights for which the *linear* corner pencil has no eigenvalue with $\Re$-part in $[\beta_\star-\delta,\beta_\star]$ (such a δ-band exists because $\beta_\star$ avoids the discrete corner spectrum, hence sits at a positive distance δ_0 from it). Choosing $\delta < \delta_0$, the weight $\beta_\star - \delta$ lies in the same admissible interval as $\beta_\star$, the two Kondratiev norms $V^{1,2}_{\beta_\star}$ and $V^{1,2}_{\beta_\star-\delta}$ are comparable on A_* (the bounded annulus excludes the vertex, where alone they would differ), and (33) follows with $\Lambda = \Lambda(\delta)$. The point is that the logarithmic growth does not move the admissible weight interval; no singular weight is "absorbed" into $\boldsymbol{v}$.

Gårding inequality. By Lemma 3 the symbol $\mathbb{L}(\boldsymbol{x})$ is strongly elliptic on A_* with the constant Legendre–Hadamard modulus μ_e. Freezing the (measurable, uniformly elliptic) coefficients on a dyadic partition $\{r \sim 2^{-j}\} \cap A_*$ and applying the constant-coefficient Gårding inequality on each annulus—valid because the principal symbol is positive definite on the Legendre–Hadamard cone—yields, after summation against the weight $r^{2\beta_\star}$,

$$a(\boldsymbol{v},\boldsymbol{v}) \geq \mu_e\, c_K^{-1}\int_{A_*} r^{2\beta_\star}|\nabla\boldsymbol{v}|^2\,dV - C\int_{A_*} r^{2\beta_\star}|\boldsymbol{v}|^2\,dV,$$

where the Korn inequality (31) converts ellipticity of the symmetric part into control of the full gradient. The remainder term has a strictly higher power of r than the Hardy weight $r^{2\beta_\star-2}$, hence by (30) it is relatively compact with respect to $\|\cdot\|_{V^{1,2}_{\beta_\star}}$; this is the compact form k, and we obtain (34) with $\gamma_1 = \mu_e\, c_K^{-1}$.

Fredholm property and exponents. Estimates (33)–(34) place the operator in the Kondratiev–Maz'ya framework [6]: it is Fredholm of index zero in $V^{1,2}_{\beta_\star}(A_*)$ whenever $\beta_\star$ avoids the spectrum of the corner operator pencil. Because the nonlinear correction to the principal symbol is subordinate (it is $o(r^{-\delta})$ for every δ, by Lemma 2), the pencil coincides with that of linearised elasticity; its eigenvalues are exactly the exponents α of (13), which are therefore unchanged by the nonlinearity. □ □

Remark 2 (The dilation core). For $f_{\log}$ the conclusion of Theorem 1 is genuinely confined to A_*: the vertex itself sits inside the core $\{r \leq r_*\}$, where $J > J_*$ and $f_{\log}$ is not strongly elliptic. Three readings are consistent with the analysis. (i) *Mild loading*: for stress-intensity $|K|$ small the core radius $r_* = (|K|/J_*)^{1/(1-\alpha)}$ is so small that the corner is elliptic on all of B_ρ down to mesh scale, and the theorem governs the entire computationally resolved region. (ii) *Sharper model statement*: the tensile corner "passes like linear elasticity" on A_*, and the only obstruction is the material law's own loss of ellipticity past J_*—a property of $f_{\log}$, not of the corner. (iii) *Open inner problem*: a complete vertex theory for $f_{\log}$ requires a separate analysis inside the non-elliptic core, which we leave open; a globally convex-volumetric barrier ($f'' \geq 0$, $r_* = 0$) removes the issue entirely and recovers the unrestricted statement. Physically, the loss of ellipticity past J_* is an artefact of the unbounded dilation that $f_{\log}$ permits: real polymer networks reach *finite extensibility*, and a volumetric law that caps the dilation, $J < J_{\max}$ (the bulk analogue of the Gent deviatoric model), with $J_{\max} \leq J_*$ keeps $f'' > 0$ throughout the attainable range and so eliminates the core. The dilation core is thus a feature of the specific large-J tail of $f_{\log}$, not a physical catastrophe.

Remark 3 (Stress is singular, operator is not). The first Piola–Kirchhoff stress still blows up, $\boldsymbol{P} \sim \mu_e\boldsymbol{F} \to \infty$ as $r \to 0$, so the tensile corner is a genuine *stress* singularity. Theorem 1 asserts only that, on the annulus A_*, this singularity lives entirely in the *right-hand side* (the residual) of the Newton step, not in its operator: under a mild volumetric penalty the nonlinear problem inherits the regularity theory of the linear one, standard graded meshes restore optimal convergence, and no structural instability occurs. This is the precise sense in which a non-compressive corner "passes like linear elasticity" away from the dilation core.

Remark 4 (The quadratic penalty does *not* pass like linear elasticity). For f_{quad}, Assumption 5 fails: by (25) and (26) the tangent grows algebraically, $\|\mathbb{L}\| = \mathcal{O}(r^{-2(1-\alpha)})$, *even in pure tension.* The coefficients of (32) are then genuinely singular, of severity comparable to the compressive Scenario B, the boundedness step above fails for $\beta_\star$, and the admissible weight interval—hence the effective corner exponents seen by the nonlinear operator—shifts relative to linear elasticity. A sharp analysis of this case (the correct weight, and whether coercivity survives) is governed by the volumetric blow-up rather than by the deviatoric singularity and is left open here.

We emphasise that this serves as a *cautionary comparator* rather than a model we advocate: as written, $f'_{\text{quad}}(1) = 0 \neq -\mu_e$, so $\frac{\kappa}{2}(J-1)^2$ does not combine with the deviatoric term $\frac{\mu_e}{2}(\mathrm{tr}(\boldsymbol{F}^T\boldsymbol{F}) - d)$ to leave the reference configuration stress-free, and in practice it is used together with an isochoric (deviatoric/volumetric) split of $\boldsymbol{F}$. The point is that a standard-looking polynomial penalty already forfeits the benign tensile behaviour enjoyed by the logarithmic model (8) actually used in [9].

Remark 5 (Self-consistency of (NC)). Hypothesis (NC) is compatible with the singular kinematics (14): an angular profile $\boldsymbol{\Phi}$ corresponding to opening/shear produces $\nabla\boldsymbol{u}$ whose dominant eigenvalue is positive (radial extension), so $\lambda_{\max} \to \infty$ while $\lambda_{\min} = \det \boldsymbol{F}/\lambda_{\max}$ need not approach zero. The complementary profile, in which the dominant eigenvalue of $\nabla\boldsymbol{u}$ is negative, drives $\lambda_{\min} \to 0$ and violates (NC); that is exactly Scenario B below.

3.6 Scenario B: the compressive singularity (the pathological case)

Now assume the far-field loading pushes the material *into* the corner ($K < 0$, radial compression).

Kinematic consequence. The naive linear prediction gives $\nabla\boldsymbol{u}$ a negative dominant eigenvalue scaling as $-|K|r^{\alpha-1}$. Because $\boldsymbol{F} = \boldsymbol{I} + \nabla\boldsymbol{u}$, as $r \to 0$ (or as the pseudo-time load factor t increases) there is a critical threshold where the local contraction annihilates the material volume:

$$\det \boldsymbol{F} \to 0 \implies J \to 0, \qquad \lambda_{\min} \to 0. \tag{35}$$

Operator blow-up (and its model dependence). Here it is the *inverse channel* of (20) that fires: $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1} \to \infty$. The dilation channel $\Theta(J)$ is now probed at $J \to 0^+$ rather than $J \to \infty$, and—exactly as in tension, but with the roles reversed—its behaviour discriminates between the two penalties. Writing $J = \prod_i \lambda_i \sim \lambda_{\min}$ near the corner (one collapsing stretch, the remaining $d-1$ bounded, cf. §3.4),

$$\Theta_{\log}(J) \sim 2\lambda_e \left|\log J\right| \to \infty, \qquad \Theta_{\text{quad}}(J) \sim 2\kappa J \to 0 \qquad (J \to 0^+), \tag{36}$$

so that the tangent norm $\|\mathbb{L}\| \le \mu_e + \Theta(J)\|\boldsymbol{F}^{-1}\|^2$ behaves as

$$\|\mathbb{L}(\boldsymbol{x})\| \sim \begin{cases} \mathcal{O}\big(\lambda_{\min}^{-2} \log \frac{1}{\lambda_{\min}}\big), & f = f_{\log}, \\ \mathcal{O}\big(\lambda_{\min}^{-1}\big), & f = f_{\text{quad}}. \end{cases} \tag{37}$$

Thus the logarithmic penalty—benign under tension—produces here the *stronger* blow-up, $\lambda_{\min}^{-2}$ *amplified* by a logarithm, whereas the quadratic penalty gives the milder rate $\lambda_{\min}^{-1}$. This reversal is not a redemption of the quadratic model: $f_{\text{quad}}(J) \to \frac{\kappa}{2} < \infty$ as $J \to 0^+$, so it provides *no energy barrier against volume collapse* and violates Assumption 1; the milder tangent growth merely reflects this missing barrier (the material is free to interpenetrate at finite energy), which is a worse pathology, not a better one. The logarithmic penalty's stronger blow-up is, conversely, the analytic price of a genuine barrier $f_{\log}(J) \to +\infty$—and it is precisely that barrier which forces, and gives physical meaning to, the bifurcation rescue of Section 4.

Ellipticity is *not* lost — the instability is structural. It is tempting to attribute the breakdown to a loss of strong ellipticity (an indefinite acoustic tensor). For this neo-Hookean model that is *false*, and it is worth stating the fact precisely, as it shapes the rest of the analysis.

Lemma 5 (Compressive strong ellipticity of the neo-Hookean tangent)**.** *For the energy* (7)*, the rank-one form of the tangent is exactly* (18)*,* $L_{iJkL}m_iN_Jm_kN_L = \mu_e|\boldsymbol{m}|^2|\boldsymbol{N}|^2 + f''(J)J^2(\boldsymbol{m}\cdot\boldsymbol{F}^{-T}\boldsymbol{N})^2$*. Wherever* $f''(J) \ge 0$ *— in particular throughout compression,* $J \le 1$*, for both penalties* (8) *— the tangent is strongly elliptic with constant* $\ge \mu_e > 0$*, uniformly in* $\boldsymbol{F}$*, and the acoustic tensor* $Q_{ik}(\boldsymbol{N}) = L_{iJkL}N_JN_L$ *is positive definite and never degenerates under compression.*

Proof. The identity is (18) (Lemma 1). As $J \to 0^+$, $f''(J)J^2 \to +\infty$ for $f_{\log}$ (cf. (56)) and $f''(J)J^2 = \kappa J^2 \geq 0$ for f_{quad}; in both cases the second term only *adds* to the constant μ_e. □ □

Thus the compressive corner is *not* a material instability: the volumetric stiffening that blows up $\|\mathbb{L}\|$ is exactly the term $f''(J)J^2$ in (18), and it is positive — it strengthens ellipticity while destroying conditioning. What fails is the *boundedness* of the operator, not its positivity.

Conclusion for Scenario B. The assumption that the deformation smoothly follows the fundamental path into the corner therefore produces, not a loss of ellipticity, but two distinct effects: (i) a blow-up of the tangent *norm*, $\|\mathbb{L}\| \to \infty$ (eq. (37)), which renders the Newton step catastrophically ill-conditioned even though it remains elliptic; and (ii) for a barrier penalty $f_{\log}$, an unphysical infinite accumulation of energy ($f(J) \to \infty$ as $J \to 0$), signalling that the smooth fundamental path is not the physical solution. The genuine mechanism that intervenes is a *structural* bifurcation of the compressed state — the corner analogue of *creasing*, an instability of a material that remains everywhere strongly elliptic (Lemma 5), driven by the geometry and boundary conditions rather than by a change of type of the PDE. To capture it we introduce in Section 4 an imperfect perturbation that regularises the bifurcation and bounds the singularity; we work from here on with the barrier model $f_{\log}$, for which collapse is genuinely resisted.

4 Imperfect bifurcation analysis and asymptotic bounds

The previous section showed that the perfect compressive path drives $\lambda_{\min} \to 0$ and blows up the tangent norm while the material remains strongly elliptic (Lemma 5), so the breakdown is a structural instability rather than a loss of type. The premise of this section is that this scenario is an idealisation: a real system always carries small disturbances—geometric imperfections, load eccentricities, material inhomogeneity—and these select a nearby *regular* solution branch on which the collapse never quite happens. We model the disturbance by a single scalar amplitude ϵ, carry out an imperfect-bifurcation analysis, and quantify how the tangent blows up as $\epsilon \to 0$. The punchline is that the blow-up is confined to a vanishingly small neighbourhood of the corner and proceeds at a controlled algebraic rate, so that for every $\epsilon > 0$ the problem is regular.

4.1 The regularised problem and the fundamental path

We localise to the corner neighbourhood B_ρ of §3, regard the loading as transmitted through data on the outer arc $\partial B_\rho \setminus (\Gamma_D \cup \Gamma_N)$, and introduce a pseudo-time load parameter $t \in [0, 1]$ that monotonically scales the compressive part of that data. Following the answer to the question posed in Scenario B, we add a small, fixed volumetric body force $\epsilon\, \boldsymbol{p}(\boldsymbol{x})$ with $\|\boldsymbol{p}\|_{L^2(B_\rho)} = 1$ and $\epsilon \ll 1$ as the imperfection (we write the field as $\boldsymbol{p}$ to avoid collision with the penalty $f(J)$). The regularised equilibrium is $G(\boldsymbol{u}, t, \epsilon) = 0$,

$$\langle G(\boldsymbol{u}, t, \epsilon), \boldsymbol{v}\rangle = \int_{B_\rho} \boldsymbol{P}(\boldsymbol{F}) : \nabla \boldsymbol{v}\, dV - \epsilon \int_{B_\rho} \boldsymbol{p} \cdot \boldsymbol{v}\, dV - \ell_t(\boldsymbol{v}) = 0 \qquad \forall \boldsymbol{v} \in V, \tag{38}$$

with $\boldsymbol{P} = \partial\Psi/\partial\boldsymbol{F}$, V the admissible test space, and ℓ_t the (linear) load functional at pseudo-time t. Let $\boldsymbol{u}_0(t)$ denote the *fundamental path*, i.e. the branch of (38) with $\epsilon = 0$ that continues the low-load compressive solution. We use the barrier penalty $f_{\log}$ throughout, so the collapse $J \to 0$ is genuinely resisted (Assumption 1); along $\boldsymbol{u}_0(t)$ the corner stretch $\lambda_c(t) := \lambda_{\min}(\boldsymbol{F}_0(t))$ decreases as $t \uparrow 1$ but, on the bifurcating branch we construct, never reaches zero.

Remark 6 (The instability is structural, not material). By Lemma 5 the tangent stays strongly elliptic throughout compression, so the bifurcation here is a *structural* one (a buckling of the boundary-value problem), not a material change of type. We work with the global form on the bounded domain B_ρ. The operator $\mathcal{L}(t) = \partial_{\boldsymbol{u}} G(\boldsymbol{u}_0(t), t, 0)$ has discrete spectrum and attains $\mu_1(t)$ as a genuine eigenvalue *provided* the fundamental stretch stays bounded away from collapse up to the crossing, $\inf_{B_\rho} \lambda_{\min}(\boldsymbol{F}_0(t)) \geq c > 0$ for $t \leq t_c$: then $\mathcal{L}(t)$ has *bounded* strongly-elliptic coefficients on the bounded domain, hence (Gårding, Korn and the Rellich embedding) compact resolvent — we make this precise as Lemma 14 in the concrete setting. The reservation is essential: what makes the problem delicate is not a loss of ellipticity but the *norm* blow-up $\|\mathbb{L}\| \to \infty$ (eq. (37)), and it is exactly this blow-up, through the scale-invariant (creasing) corner, that would push the bottom of the spectrum into the *essential* spectrum and destroy attainment. Crucially, that blow-up occurs only on the unphysical continuation past t_c, where $\lambda_{\min} \to 0$; at the crossing itself the spectrum is discrete *so long as buckling precedes collapse*. The coercivity constant thus stays positive while the boundedness constant diverges, and the crossing eigenvalue $\mu_1(t)$ and the conditioning are controlled by different mechanisms. Assumption 6 records the spectral facts (attainment and simplicity) we use; §5.8 shows that, for the concrete model, both follow once buckling is known to precede collapse.

4.2 What is provable, and what must be assumed

It is worth separating the genuinely provable facts about the critical point from the irreducible hypotheses. Let $Q_t(\boldsymbol{v}) := \int_{B_\rho} \nabla \boldsymbol{v} : \mathbb{L}(\boldsymbol{F}_0(t)) : \nabla \boldsymbol{v}\, dV$ be the second-variation form on the fundamental path and $\mu_1(t) := \inf_{\|\boldsymbol{v}\|_V = 1} Q_t(\boldsymbol{v})$.

The critical load exists (conditionally provable). A clarification is essential here. By Lemma 5 the neo-Hookean material remains strongly elliptic throughout compression, so $\mu_1(t)$ *cannot* be driven negative by a rank-one (material) instability: any crossing must be a *structural* loss of positivity of the global form Q_t, i.e. a buckling of the boundary-value problem in which the destabilising test field is not rank-one. This is the creasing mechanism.

Lemma 6 (Existence of the critical load). *Assume (i) coercivity at zero load, $\mu_1(0) > 0$, and (ii) a structural instability of the compressed state: there exist $t_* \in (0, 1)$ and $\boldsymbol{v}_* \in V$ with $Q_{t_*}(\boldsymbol{v}_*) < 0$. Then μ_1 is continuous on $[0, t_*]$ and there is a first $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$ and $\mu_1(t) > 0$ for $t < t_c$.*

Proof. At $t = 0$, $\boldsymbol{F}_0 = \boldsymbol{I}$ and $\mathbb{L} = \mu_e \mathbb{I} +$ (volumetric at J=1) is positive definite, so with Korn's inequality $\mu_1(0) > 0$. Hypothesis (ii) gives $\mu_1(t_*) \leq Q_{t_*}(\boldsymbol{v}_*)/\|\boldsymbol{v}_*\|_V^2 < 0$. As $t \mapsto \mathbb{L}(\boldsymbol{F}_0(t))$ is continuous, so is μ_1, and the intermediate value theorem gives the first zero t_c. □ □

Hypothesis (i) is provable for neo-Hookean (positive-definiteness at $\boldsymbol{F} = \boldsymbol{I}$ plus Korn). Hypothesis (ii) — the existence of a structurally destabilising field — is the genuine content and is *not* elementary: it is the corner counterpart of the Biot/creasing instability [1, 3] and we do not establish it here.

The critical mode (assumed).

Assumption 6 (Critical mode). At t_c the infimum $\mu_1(t_c) = 0$ is attained at a mode $\boldsymbol{\phi} \in V$, $\|\boldsymbol{\phi}\|_V = 1$, spanning a one-dimensional, isolated kernel of $\mathcal{L}_c$ (the rest of the spectrum bounded away from 0), with $\boldsymbol{\phi}$ localised near the corner.

This bundles two spectral facts: *attainment* (the minimising sequence does not escape into the corner as essential spectrum) and *simplicity*. Simplicity is generic and, for a symmetric corner, can be argued from a reflection symmetry about the bisector; attainment is the borderline-elliptic difficulty flagged in Remark 6. Neither is irreducible: in the concrete setting both are *proved* (§5.5, Lemma 14), the only residue being the cleaner geometric condition that buckling precede pointwise collapse.

The crossing is transversal (provable under monotone loading).

Lemma 7 (Transversality)**.** *Suppose the loading is monotonically destabilising at t_c,*

$$\langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle := \int_{B_\rho} \nabla\boldsymbol{\phi} : \partial_t \mathbb{L}(\boldsymbol{F}_0(t_c)) : \nabla\boldsymbol{\phi}\, dV < 0. \tag{M}$$

Then, under Assumption 6, μ_1 is differentiable at t_c with $a := -\mu_1'(t_c) = -\langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle > 0$.

Proof. For a simple eigenvalue, first-order perturbation theory gives $\mu_1'(t_c) = \langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle$; hypothesis (M) fixes the sign. □ □

Condition (M) is the precise meaning of "more compression ⇒ less stiffness in the buckling direction" and holds for any monotone compressive load path; it is an input on the *loading*, not on the solution.

The imperfection is active (provable, generic).

Lemma 8 (Generic activation)**.** *For every $\boldsymbol{p}$ outside the closed codimension-one subspace $\boldsymbol{\phi}^\perp$, the projection $\eta := \int_{B_\rho} \boldsymbol{p}\cdot\boldsymbol{\phi}\, dV \neq 0$. Hence $\eta \neq 0$ for generic imperfection.*

Remark 7 (Universality of the imperfection). The volumetric body force $\epsilon\,\boldsymbol{p}$ is chosen only for concreteness; the analysis depends on the imperfection *solely* through the scalar $\eta = \langle \boldsymbol{p}, \boldsymbol{\phi}\rangle$, so any perturbation with a nonzero $\boldsymbol{\phi}$-projection produces the identical reduced equation (41). Indeed, write a generic imperfect equilibrium as $G(\boldsymbol{u}, t, 0) = \epsilon\,\boldsymbol{h}$ with $\boldsymbol{h} \in V'$ the (linearised) imperfection functional. A boundary-traction defect on Γ_N contributes $\boldsymbol{h}(\boldsymbol{v}) = \int_{\Gamma_N} \boldsymbol{t}_1 \cdot \boldsymbol{v}\, dS$; a geometric imperfection — rounding the corner with radius $\sim \epsilon$, or a small shape variation $\boldsymbol{x} \mapsto \boldsymbol{x} + \epsilon\,\boldsymbol{\theta}(\boldsymbol{x})$ — enters, after pulling back to the reference domain, as a fixed load functional $\boldsymbol{h}(\boldsymbol{v}) = \int_{B_\rho} \mathbb{S}[\boldsymbol{\theta}] : \nabla\boldsymbol{v}\, dV$ linear in the shape field $\boldsymbol{\theta}$. In every case the kernel projection $\langle G, \boldsymbol{\phi}\rangle = 0$ yields the same normal form with η replaced by $\langle \boldsymbol{h}, \boldsymbol{\phi}\rangle$, nonzero for $\boldsymbol{h}$ outside the closed codimension-one subspace $\boldsymbol{\phi}^\perp$ (Lemma 8). The Koiter $\frac{2}{3}$ law (44) and all subsequent bounds are therefore universal across generic imperfections, not an artefact of the body-force model.

The nonlinearity is explicit; only the sign of its orthogonal-mode feedback is open. For the neo-Hookean model the reduced coefficients are not opaque. By Lemma 13 the deviatoric energy is quadratic and drops out, so every coefficient of order ≥ 3 is purely volumetric; moreover the quadratic coefficient $c_2 := \frac{1}{2}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^2 G[\boldsymbol{\phi}, \boldsymbol{\phi}]\rangle$ vanishes *identically* for a mode antisymmetric under the lateral, face-preserving reflection R of §5.5 — not the corner bisector $\theta \mapsto \frac{\pi}{2} - \theta$, which swaps Γ_D and Γ_N and is no symmetry at all. What these structural facts do *not* settle is the *sign* of the cubic and quintic coefficients; indeed, by Lemma 13(ii), the explicit single-mode (frozen) parts point *against* subcriticality, $b_{\mathrm{dir}} > 0$ and $d_{\mathrm{dir}} < 0$ in compression. The subcritical pattern therefore hinges on the geometry-dependent orthogonal-mode feedback overturning these signs; we record it as a residual hypothesis to be verified numerically (Remark 8) rather than asserted from the creasing analogy [1, 3].

Assumption 7 (Subcritical sign pattern (residual, geometry-dependent))**.** With $c_2 = 0$ by Lemma 13(i), the full reduced cubic and quintic coefficients

$$b := \tfrac{1}{6}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^3 G[\boldsymbol{\phi}, \boldsymbol{\phi}, \boldsymbol{\phi}]\rangle, \qquad d := \tfrac{1}{120}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^5 G[\boldsymbol{\phi}, \dots, \boldsymbol{\phi}]\rangle, \tag{39}$$

each the frozen part of Lemma 13 corrected by the orthogonal-mode feedback, satisfy $b < 0 < d$ (subcritical with quintic saturation). This is a condition on the corner geometry, not on the material model.

Remark 8 (Status of the subcritical signs: geometry-decided, checked numerically)*.* Assumption 7 is the one genuinely residual hypothesis of the construction, and Lemma 13 localises exactly where its content lies. The deviatoric response and the quadratic coefficient are settled ($c_2 = 0$, no feedback correction); the cubic and quintic decompose as $b = b_{\mathrm{dir}} + b_\psi$ and $d = d_{\mathrm{dir}} + d_\psi$, with the single-mode parts explicit and, in compression, of definite but *adverse* sign ($b_{\mathrm{dir}} = 2\int_{B_\rho} f''(J_0) d_\phi^2 \, dV > 0$, $d_{\mathrm{dir}} = \int_{B_\rho} f'''(J_0) d_\phi^3 \, dV < 0$). The corrections b_ψ, d_ψ are the Lyapunov–Schmidt feedback through the orthogonal complement $V^\perp$: quadratic forms in the second-order field $\boldsymbol{\psi}$ that depend on the opening angle, the Dirichlet–Neumann arrangement and the mode $\boldsymbol{\phi}$ — on the geometry, not on f. Subcriticality thus requires b_ψ to overturn the explicit positive b_{dir} (and likewise for d). We do not sign b_ψ, d_ψ analytically; they are read off directly from the finite-element eigenmode and reduced operator of the cut-FEM study [9], the proper arbiter of a geometry-dependent sign. Were the verdict $b > 0$ instead, the bifurcation would be supercritical and *more* benign — the physical branch smooth through t_c with no snap at onset — and the regularisation conclusions of §5.8 and Proposition 4 would hold a fortiori; the complementary transcritical case $c_2 \neq 0$ is covered in Remark 9.

4.3 Lyapunov–Schmidt reduction

Proposition 1 (Normal form)**.** *Under Assumption 6, Lemma 7 and Assumption 7, there is a neighbourhood of* $(\boldsymbol{u}_0(t_c), t_c, 0)$ *in which the solutions of* (38) *are exactly*

$$\boldsymbol{u}(t, \epsilon) = \boldsymbol{u}_0(t) + w\,\boldsymbol{\phi} + \boldsymbol{\psi}(w, t, \epsilon), \qquad \boldsymbol{\psi} \in V^\perp := \{\boldsymbol{\phi}\}^\perp, \tag{40}$$

where $\boldsymbol{\psi}$ *is smooth with* $\boldsymbol{\psi} = \mathcal{O}(w^2 + \epsilon)$*, and the amplitude* w *solves the scalar imperfect bifurcation equation, retained to quintic order because the cubic is destabilising (*$b < 0$*),*

$$a\,(t_c - t)\,w + b\,w^3 + d\,w^5 = \epsilon\,\eta + \mathcal{O}\big(\epsilon^2 + \epsilon w + |t_c - t|\,w^2 + w^6\big), \tag{41}$$

with $a > 0$ *(Lemma 7),* $b < 0 < d$ *(Assumption 7) and* η *as in Lemma 8.*

Proof. Let P be the V-orthogonal projection onto $\operatorname{span}\{\boldsymbol{\phi}\}$ and $Q = I - P$. Write $\boldsymbol{u} = \boldsymbol{u}_0(t) + w\boldsymbol{\phi} + \boldsymbol{\psi}$ with $\boldsymbol{\psi} \in V^\perp$. The range equation $Q\,G = 0$ has invertible linearisation $Q\,\mathcal{L}_c\,Q$ on $V^\perp$ by Assumption 6, so the implicit function theorem yields a unique smooth $\boldsymbol{\psi} = \boldsymbol{\psi}(w, t, \epsilon)$ with $\boldsymbol{\psi}(0, t_c, 0) = 0$ and $\boldsymbol{\psi} = \mathcal{O}(w^2 + \epsilon + |t_c - t|)$. Substituting into the kernel equation $\langle G, \boldsymbol{\phi}\rangle = 0$ and Taylor-expanding G about $(\boldsymbol{u}_0(t_c), t_c, 0)$ gives (41): the linear term $a(t_c - t)w$ from Lemma 7, the cubic bw^3 and quintic dw^5 from Assumption 7 (the quadratic vanishing by symmetry), and the inhomogeneity $\epsilon\eta$ from $-\epsilon\langle \boldsymbol{p}, \boldsymbol{\phi}\rangle$. Because $b < 0$, the cubic alone does not saturate the amplitude, which is why the quintic dw^5 ($d > 0$) is retained. □ □

4.4 Subcritical branch structure and imperfection sensitivity

Perfect system. Set $\epsilon = 0$ and $t < t_c$. Dividing (41) by w gives $a(t_c - t) + bw^2 + dw^4 = 0$, i.e.

$$w^2 = \frac{-b \pm \sqrt{b^2 - 4ad(t_c - t)}}{2d}, \tag{42}$$

which (with $a > 0$, $b < 0$, $d > 0$) admits real nontrivial branches only for $t \geq t_L$, where

$$t_L := t_c - \frac{b^2}{4ad} < t_c, \qquad w_L^2 = \frac{-b}{2d} = \mathcal{O}(1). \tag{43}$$

At t_L the two branches meet in a limit point (fold) at the *finite* amplitude w_L. The fundamental branch $w = 0$ is stable for $t < t_c$ and unstable for $t > t_c$; the upper nontrivial branch is stable. Hence on $[t_L, t_c)$ the compressed state and a finite-amplitude creased state coexist: the bifurcation is subcritical, with hysteresis and a *snap* at t_c.

Imperfect system.

Proposition 2 (Imperfection sensitivity, Koiter $\frac{2}{3}$ law)**.** *For $\epsilon > 0$ the compressed branch terminates at a limit point that issues from the bifurcation point $(0, t_c)$ and is depressed below t_c according to*

$$t_s(\epsilon) = t_c - C\,|\eta\,\epsilon|^{2/3} + o(\epsilon^{2/3}), \qquad C = \frac{3\,|b|^{1/3}}{2^{2/3}\,a} > 0, \tag{44}$$

so the load at which the compressed branch ceases to exist is reduced by $\mathcal{O}(\epsilon^{2/3})$. The turning amplitude there is itself small, $w_* = \mathcal{O}(\epsilon^{1/3})$; the $\mathcal{O}(1)$ amplitude is that of the creased branch onto which the structure then snaps.

Proof. Write (41) as $g(w, t) = 0$ with

$$g(w, t) := a(t_c - t)\,w + b\,w^3 + d\,w^5 - \epsilon\eta,$$

the analytic remainder of (41) being subdominant at the scaling found below (we verify this at the end). A limit point in the load is a point of the solution set at which the branch turns, i.e. at which t is stationary along the curve; differentiating $g = 0$ implicitly, this is the pair

$$g(w, t) = 0, \qquad g_w(w, t) = a(t_c - t) + 3b\,w^2 + 5d\,w^4 = 0. \tag{45}$$

We follow the limit point that issues from the bifurcation point $(0, t_c)$ as $\epsilon \to 0$; this is the one that caps the slightly deflected compressed branch and hence sets the snap load.

Turning amplitude. Using the second equation of (45) to eliminate the load, $a(t_c - t) = -(3b\,w^2 + 5d\,w^4)$, and substituting into the first gives a single equation for the turning amplitude,

$$-2b\,w^3 - 4d\,w^5 = \epsilon\eta. \tag{46}$$

Because $b < 0$, the small root of (46) as $\epsilon \to 0$ obeys $-2b\,w^3 = \epsilon\eta\,(1 + o(1))$, the quintic entering only at relative order w^2; hence

$$w_*(\epsilon) = \left(\tfrac{|\eta|\,\epsilon}{2|b|}\right)^{1/3} (1 + o(1)) = \mathcal{O}(\epsilon^{1/3}), \tag{47}$$

in agreement with the unstable-branch scaling (48) evaluated at the turn. In particular $w_* \to 0$, so at this limit point the cubic dominates the quintic, $d\,w_*^4/(b\,w_*^2) = \mathcal{O}(\epsilon^{2/3})$.

Snap load. Inserting (47) into $g_w = 0$ and discarding the quintic, a relative $\mathcal{O}(\epsilon^{2/3})$ correction,

$$a\big(t_c - t_s(\epsilon)\big) = -3b\,w_*^2\,(1+o(1)) = 3|b|\Big(\tfrac{|\eta|\,\epsilon}{2|b|}\Big)^{2/3}(1+o(1)),$$

which is (44) with $C = 3|b|^{1/3}/(2^{2/3}a) > 0$, the classical Koiter $\frac{2}{3}$ power law [5, 2].

Remainder. At the limit point $w = \mathcal{O}(\epsilon^{1/3})$ and $t_c - t = \mathcal{O}(\epsilon^{2/3})$, so the omitted terms $\mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t|\,w^2 + w^6)$ of (41) are $\mathcal{O}(\epsilon^{4/3})$, a relative $\mathcal{O}(\epsilon^{1/3})$ against the dominant balance $a(t_c - t)w \sim b\,w^3 \sim \epsilon\eta = \mathcal{O}(\epsilon)$. They perturb w_* and $t_s(\epsilon)$ only within the stated $o(\epsilon^{2/3})$, leaving the exponent and C unchanged. □ □

The unstable continuation branch. Separately, near t_c the small-amplitude (backward-bending, unstable) branch of (41) satisfies, at $t = t_c$, $b\,w^3 = \epsilon\eta\,(1+o(1))$ with the quintic subdominant, so

$$|w(\epsilon)| = \Big(\frac{|\eta|}{|b|}\Big)^{1/3}\epsilon^{1/3}\,(1+o(1)), \qquad \epsilon \to 0. \tag{48}$$

This is the branch a displacement- or arc-length-controlled solver tracks through the instability, and it is the one that feeds the tangent estimate below; it is *not* the branch a load-controlled physical system follows.

Remark 9 (Robustness to the bifurcation type: pitchfork vs. transcritical). Assumption 7 takes the reduced potential to be odd ($c_2 = 0$), a subcritical *pitchfork*; by the discussion there this holds when the critical mode is antisymmetric under the lateral reflection R of §5.5. If that symmetry is absent or the ground mode lies in the symmetric sector, the quadratic coefficient $c_2 \neq 0$ survives and (41) is instead *transcritical*,

$$a\,(t_c - t)\,w + c_2\,w^2 + \mathcal{O}(w^3) = \epsilon\,\eta. \tag{49}$$

The qualitative architecture of the section is unaffected. Repeating the limit-point analysis of Proposition 2 on (49) (solve $g = g_w = 0$ with $g = a(t_c - t)w + c_2 w^2 - \epsilon\eta$) gives a turning amplitude $w_*^2 = -\epsilon\eta/c_2 = \mathcal{O}(\epsilon)$ and a snap load depressed by the classical Koiter *one-half* law,

$$t_s(\epsilon) = t_c - C'\,|\eta\,\epsilon|^{1/2} + o(\epsilon^{1/2}), \tag{50}$$

in place of the $\frac{2}{3}$ law and $w_* = \mathcal{O}(\epsilon^{1/3})$ of the pitchfork. The determinant relief (52)–(54) still lifts $\lambda_{\min}$ off zero, now at order $w_*^2 = \mathcal{O}(\epsilon)$ rather than $\mathcal{O}(\epsilon^{2/3})$, so the regularised tangent bound of §5.8 reads $\|\mathbb{L}\| = \mathcal{O}(\epsilon^{-2}\log\frac{1}{\epsilon})$ in place of $\mathcal{O}(\epsilon^{-4/3})$, and the localisation of Proposition 4 persists with the correspondingly shifted exponents. In particular the central result — a bounded physical branch carrying a vanishingly localised, regularised tangent blow-up for every $\epsilon > 0$ — is insensitive to whether the corner bifurcation is a subcritical pitchfork or a transcritical fold; only the imperfection-sensitivity exponents change.

4.5 Kinematic relief: lower bound on J

The heart of the regularisation is geometric: the buckling mode trades pure compression for rotation and shear, which lifts the determinant off zero. The reason the relief is second order in w—and hence why the scaling exponents below are what they are—is that the mode is, to leading order, a *rotation*, and rotations do not change volume at first order. We do *not* assume the first-order volume change vanishes exactly; we use the quantitative bound on it furnished by the energy barrier (Lemma 10) and show it is dominated by the second-order relief.

Lemma 9 (Determinant relief)**.** *Let $\boldsymbol{F} = \boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi} + \mathcal{O}(w^2)$ at the corner, with $\boldsymbol{F}_0 = \boldsymbol{F}_0(t)$ on the loaded path, $J_0 = \det \boldsymbol{F}_0$, and write the first-order volume change as*

$$D_1 := \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = J_0\, \boldsymbol{F}_0^{-T} : \nabla\boldsymbol{\phi}. \tag{51}$$

By the barrier estimate (57) of Lemma 10, the critical mode is asymptotically *volume-preserving, $|D_1| \leq c_\phi\, J_0/\sqrt{|\log J_0|}$, so D_1 does not vanish at finite load but is itself small with J_0. Let $d_\phi := \frac{1}{2}\frac{d^2}{dw^2}\big|_0 \det(\boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi}) > 0$ be the second-order coefficient of the determinant — the second mixed discriminant of $\boldsymbol{F}_0$ ($d-2$ copies) with $\nabla\boldsymbol{\phi}$ (two copies), which for $d = 2$ reduces to* $\det(\nabla\boldsymbol{\phi})$*. Then*

$$J = \det \boldsymbol{F} = J_0 + w\, D_1 + w^2\, d_\phi + \mathcal{O}(w^3), \tag{52}$$

and the linear term is dominated by the other two: for $|w| \leq w_0$ small,

$$J \;\geq\; \tfrac{1}{2}\, J_0 + \tfrac{1}{2}\, w^2\, d_\phi. \tag{53}$$

Consequently, with $\lambda_c = \lambda_{\min}(\boldsymbol{F}_0)$ and the remaining $d-1$ stretches of $\boldsymbol{F}$ bounded, $\prod_{i\geq 2} \lambda_i(\boldsymbol{F}) = \mathcal{O}(1)$,

$$\lambda_{\min}(\boldsymbol{F}) = \frac{J}{\prod_{i\geq 2}\lambda_i(\boldsymbol{F})} \;\gtrsim\; C_1\lambda_c + C_2\, w^2, \qquad C_1, C_2 > 0. \tag{54}$$

Proof. Expanding the determinant in w, $\det(\boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi}) = \sum_{k=0}^{d} w^k\, D_k(\boldsymbol{F}_0, \nabla\boldsymbol{\phi})$ with D_k the k-th mixed discriminant; $D_0 = \det \boldsymbol{F}_0 = J_0$, $D_1 = \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi}$, and $D_2 = d_\phi$, giving (52) (the $\mathcal{O}(w^3)$ collecting $D_3, \ldots, D_d$). For (53) we bound the linear term two ways using $|D_1| \leq c_\phi J_0/\sqrt{|\log J_0|}$. By Young's inequality $|w\, D_1| \leq \frac{1}{2} d_\phi w^2 + \frac{1}{2d_\phi} D_1^2$, and $D_1^2 \leq c_\phi^2 J_0^2/|\log J_0| \leq \frac{1}{2} d_\phi J_0$ once J_0 is small (the factor $J_0/|\log J_0| \to 0$). Hence $|w\, D_1| \leq \frac{1}{2} d_\phi w^2 + \frac{1}{2} J_0$, and inserting into (52) (absorbing the $\mathcal{O}(w^3)$ into the constants for $|w| \leq w_0$) yields (53). Thus the residual first-order term, far from dominating, is swallowed by the barrier-controlled J_0 and the relief $w^2 d_\phi$ — this is the precise content of "buckling relieves compression" at finite load, with no appeal to an exact rotation condition. Since $J = \prod_i \lambda_i$ and the $d-1$ largest stretches are $\mathcal{O}(1)$ on the relieved branch (the buckling redirects, but does not unboundedly stretch, the material near the corner), (52) gives (54). $\square$ $\square$

The estimate (54) carries two competing floors, $C_1\lambda_c$ and $C_2 w^2$, and which one binds depends on the load. At the bifurcation point itself the perfect stretch $\lambda_c(t_c) > 0$ is finite (buckling precedes collapse, Remark 6); there $C_1\lambda_c$ dominates, $\lambda_{\min}$ is bounded below by a constant, and *no* blow-up occurs — consistently with the bounded-tangent physical branch of Proposition 3. The tangent divergence is felt only along the *unstable continuation* that a solver tracks to deeper loads, where the perfect path drives $\lambda_c(t) \to 0$; precisely in the regime $\lambda_c \lesssim w^2$ the relief term takes over. Combining (54) with the branch amplitude $w \sim \epsilon^{1/3}$ of (48) then gives, in that near-collapse regime,

$$\lambda_{\min}(\epsilon) \gtrsim C_2\, w(\epsilon)^2 \sim \mathcal{O}(\epsilon^{2/3}), \qquad \|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1} \sim \mathcal{O}(\epsilon^{-2/3}). \tag{55}$$

Outside that regime ($\lambda_c \gtrsim \epsilon^{2/3}$) the floor $C_1\lambda_c$ keeps $\|\boldsymbol{F}^{-1}\|$ even smaller, so (55) is the worst case. On the stable physical branch the picture is milder still; we contrast the two after Theorem 2.

The smallness of the first-order volume change D_1 in (51) is not an extra assumption: it is *forced*, quantitatively, by the energy barrier, as we now show. This is the precise content of the informal statement "buckling relieves compression"—the barrier penalises any finite-energy mode that changes volume at first order, driving $D_1 \to 0$ with J_0 at the rate (57), which (Lemma 9) is fast enough for the second-order relief to dominate.

Lemma 10 (The barrier forces the rotation condition)**.** *Let $f = f_{\log}$. Then the bulk modulus diverges at collapse,*

$$f''(J) = \frac{\mu_e + 2\lambda_e - 2\lambda_e \log J}{J^2} \sim \frac{2\lambda_e\,|\log J|}{J^2} \to +\infty \qquad (J \to 0^+), \tag{56}$$

and consequently any mode $\boldsymbol{\phi}$ of bounded second-variation energy $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi} \rangle \le C$ on the near-collapse fundamental state $\boldsymbol{F}_0$ satisfies

$$\operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = \mathcal{O}\Big(\frac{J_0}{\sqrt{|\log J_0|}}\Big) \to 0 \qquad (J_0 \to 0^+). \tag{57}$$

In particular the critical buckling mode is first-order volume-preserving in the collapse limit, so (51) *holds asymptotically along the loaded path, and the residual first-order term in* (52) *is dominated by the relief term $w^2 d_\phi$.*

Proof. Differentiating $f'_{\log}(J) = -\mu_e/J + 2\lambda_e \log J/J$ gives (56). The volumetric part of the second variation of Ψ in the direction $\boldsymbol{\phi}$ contains the term $f''(J_0)\,(\delta J)^2$, where $\delta J = \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = J_0\,\boldsymbol{F}_0^{-T} : \nabla\boldsymbol{\phi}$ is the first variation of the determinant. Since every term other than this one is bounded on the fundamental state, boundedness of $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi} \rangle$ forces $f''(J_0)\,(\delta J)^2 \le C'$, hence $(\delta J)^2 \le C'/f''(J_0) \sim C'\,J_0^2/(2\lambda_e|\log J_0|)$. Taking square roots yields (57). □ □

4.6 Regularised tangent bound and spatial localisation

Proposition 3 (Stable physical branch: bounded tangent)**.** *Under load control the structure follows the compressed fundamental branch up to the snap at $t_s(\epsilon) \le t_c$ (Proposition 2) and then jumps to the creased branch of amplitude $w = \mathcal{O}(1)$. On the compressed branch $\lambda_{\min} \ge \lambda_c(t) \ge \lambda_c(t_c) > 0$; after the snap the $\mathcal{O}(1)$ amplitude relieves the compression to $\lambda_{\min} = \mathcal{O}(1)$ via Lemma 9. Hence on the physical branch*

$$\max_{\boldsymbol{x}\in B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\| = \mathcal{O}(1) \quad \textit{uniformly in } \epsilon, \tag{58}$$

and the sole effect of the imperfection is the $\mathcal{O}(\epsilon^{2/3})$ reduction (44) *of the snap load. There is no tangent blow-up on the physical branch.*

Theorem 2 (Unstable continuation branch: regularised blow-up)**.** *Under Assumptions 1, 6, 7 and the conclusions* (55) *of Lemma 9, with the barrier penalty $f_{\log}$, the tangent* along the unstable continuation branch (48)*—the branch tracked by a displacement- or arc-length-controlled solver—satisfies*

$$\max_{\boldsymbol{x}\in B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\| \;\le\; C\,\Theta(J)\,\|\boldsymbol{F}^{-1}\|^2 \;\sim\; \mathcal{O}\big(\epsilon^{-4/3}\,\log\tfrac{1}{\epsilon}\big), \qquad \epsilon \to 0. \tag{59}$$

Proof. By (11), $\|\mathbb{L}\| \le \mu_e + \Theta(J)\|\boldsymbol{F}^{-1}\|^2$. For $f_{\log}$, $\Theta(J) = \mathcal{O}(\log\frac{1}{J})$ as $J \to 0$ by (36), and on the unstable branch $J \sim \lambda_{\min} \sim \epsilon^{2/3}$ by (55), so $\Theta(J) = \mathcal{O}(\log\frac{1}{\epsilon})$; multiplying by $\|\boldsymbol{F}^{-1}\|^2 \sim \epsilon^{-4/3}$ gives (59). □ □

The two results together are the precise sense in which "the singularity never occurs in reality": a load-controlled physical system snaps and keeps a bounded tangent (Proposition 3), and the divergent $\epsilon^{-4/3}$ rate is felt only by an algorithm that suppresses the snap and tracks the unstable branch through the instability (Theorem 2). The spatial localisation below applies to that branch.

The maximum is, however, achieved only in a shrinking zone, and we can now identify the localisation exponent exactly. The relevant geometry is the *collapse locus* of the perfect field.

Lemma 11 (Transversal collapse: $s = 1$)**.** *On the perfect (unphysical) continuation of the fundamental path beyond t_c, let $\Sigma_c \subset B_\rho$ be the collapse locus on which the smallest stretch vanishes, $\lambda_{\min} = 0$ (equivalently $J = 0$); it is generically of codimension one. At any point of Σ_c where $\lambda_{\min}$ is a simple singular value of $\boldsymbol{F}$, $\lambda_{\min}$ is a smooth function of $\boldsymbol{x}$, and generically it vanishes* transversally*,*

$$\lambda_{\min}^{\text{field}}(\boldsymbol{x}) = \kappa_* \operatorname{dist}(\boldsymbol{x}, \Sigma_c) + \mathcal{O}\big(\operatorname{dist}^2\big), \qquad \kappa_* > 0. \tag{60}$$

Hence the localisation exponent is $s = 1$. (A non-generic tangential contact, $\nabla\lambda_{\min} = 0$ on Σ_c, would give $s \geq 2$.)

Remark 10 (Extended arc versus point vertex). The codimension-one hypothesis is generic but not forced. In a compressive mixed corner the most severe contraction is frequently attained *exactly at the vertex* $r = 0$, in which case the collapse locus degenerates to the single point $\Sigma_c = \{0\}$ — codimension *two* in $d = 2$. The localisation conclusion then only improves: the relieved floor $\lambda_{\min} \gtrsim \max\{\kappa_* r, c\,\epsilon^{2/3}\}$ now confines the high-stiffness set to a *disc* of radius $\ell(\epsilon) \sim \epsilon^{2/3}$ rather than a tube of fixed length, so its area is $\mathcal{O}(\ell(\epsilon)^2) = \mathcal{O}(\epsilon^{4/3})$ rather than $\mathcal{O}(\epsilon^{2/3})$, and the L^q-bound of Proposition 4 acquires the stronger tube contribution $\epsilon^{-4q/3}\,\ell(\epsilon)^2 = \epsilon^{\frac{4}{3}(1-q)}$, finite for every $q < 1$ instead of $q < \frac{1}{2}$. The extended-arc estimate below is therefore the conservative case; a point vertex is strictly more integrable.

Proof. Where the smallest singular value is simple it is an analytic function of the entries of $\boldsymbol{F}$, hence smooth in $\boldsymbol{x}$. A smooth scalar field vanishing on Σ_c has, by Taylor expansion, $\lambda_{\min} = \nabla\lambda_{\min} \cdot \boldsymbol{n} \operatorname{dist}(\boldsymbol{x}, \Sigma_c) + \mathcal{O}(\operatorname{dist}^2)$ with $\boldsymbol{n}$ the unit normal; transversality $\nabla\lambda_{\min} \neq 0$ (the generic case, by Sard's theorem) gives (60) with $\kappa_* = |\nabla\lambda_{\min}| > 0$. □ □

On the imperfect branch the relief of (55) imposes the floor $\lambda_{\min}(\boldsymbol{x}, \epsilon) \gtrsim \max\{\kappa_* \operatorname{dist}(\boldsymbol{x}, \Sigma_c), c\,\epsilon^{2/3}\}$, so the high-stiffness zone is the tube of half-width $\ell(\epsilon)$ around Σ_c in which the floor is active, with (using $s = 1$)

$$\ell(\epsilon) \sim \frac{c}{\kappa_*}\,\epsilon^{2/3} \xrightarrow[\epsilon \to 0]{} 0. \tag{61}$$

Proposition 4 (Narrowness of the singular zone)**.** *With $\ell(\epsilon)$ as in* (61) *and $s = 1$ (Lemma 11), the set on which the tangent is near its maximum is a tube about Σ_c of area $\mathcal{O}(|\Sigma_c|\,\ell(\epsilon)) = \mathcal{O}(\epsilon^{2/3}) \to 0$, and the integrated stiffness obeys*

$$\int_{B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\|^q \, dV \;\leq\; C\Big(1 + \epsilon^{-4q/3}\,\ell(\epsilon)\,\log^q \tfrac{1}{\epsilon}\Big) \;=\; C\Big(1 + \epsilon^{\frac{2}{3}(1-2q)} \log^q \tfrac{1}{\epsilon}\Big), \tag{62}$$

which stays bounded as $\epsilon \to 0$ for every $q < \frac{1}{2}$. Thus, although the pointwise tangent diverges like $\epsilon^{-4/3}$, its L^q-mass over the corner remains finite for all $q < \frac{1}{2}$: the blow-up is genuinely localised to a tube of vanishing area, and for every $\epsilon > 0$ the regularised problem is well posed.

Proof. Split B_ρ into the tube $T_\epsilon = \{\operatorname{dist}(\cdot, \Sigma_c) < \ell(\epsilon)\}$ and its complement. On T_ϵ, $\|\mathbb{L}\| \sim \epsilon^{-4/3} \log \frac{1}{\epsilon}$ (Theorem 2) and $|T_\epsilon| \sim |\Sigma_c|\ell(\epsilon) \sim \epsilon^{2/3}$, giving the displayed contribution. Off T_ϵ, using the transversal coordinate $\varrho = \operatorname{dist}(\cdot, \Sigma_c)$ and $\lambda_{\min} \sim \kappa_*\varrho$, one has $\|\mathbb{L}\| \sim \varrho^{-2} \log \frac{1}{\varrho}$, and $\int_{\ell(\epsilon)}^{\rho} \varrho^{-2q} \log^q \frac{1}{\varrho}\, d\varrho$ converges as $\ell(\epsilon) \to 0$ precisely when $2q < 1$. Both regimes are controlled iff $q < \frac{1}{2}$. □ □

4.7 Discussion

The structure of the result is the one anticipated physically, and it splits cleanly along the stability of the branch followed. The idealised noiseless problem ($\epsilon = 0$) is singular—the tangent is unbounded and ellipticity is lost—but it is a measure-zero idealisation. Because the corner instability is subcritical (the creasing analogy, Assumption 7), a load-controlled physical system never traverses the singular continuation: it stays on the compressed branch and *snaps* to a finite-amplitude crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$ (Proposition 2), keeping a bounded tangent throughout (Proposition 3). The much-discussed divergence is therefore not seen by the physical system at all; it is the worst case for an algorithm—displacement or arc-length control, numerical continuation—that suppresses the snap and tracks the *unstable* branch through the instability. On that branch the determinant is still lifted off zero (Lemma 9, with the volume-preserving character of the mode *forced* by the energy barrier, Lemma 10), the tangent diverges only algebraically, $\epsilon^{-4/3}$ up to a logarithm (Theorem 2), and the singular character is confined to a tube of width $\ell(\epsilon) \sim \epsilon^{2/3} \to 0$ about the collapse locus, remaining L^q-integrable for all $q < \frac{1}{2}$ (Lemma 11, Proposition 4). That algebraic rate is the mirror image—through the inverse channel—of the benign tensile growth of §3.4; the same logarithm, traceable to the barrier $f_{\log}$, appears in both. Two practical consequences follow. First, the bifurcation is the corner-localised analogue of the classical *creasing* instability of compressed elastomers [1, 3], now seated at a geometric Dirichlet–Neumann junction rather than on a flat surface. Second, the dichotomy is a concrete warning for computation: a solver that continues through the instability will see the conditioning degrade as $\epsilon^{-4/3}$ and should either admit the snap or refine inside the tube (61); the load-control reduction (44) tells it when the snap is due. The reduction to scalar imperfect-bifurcation form follows the classical theory [2, 4, 5].

5 A concrete realization: the clamped–free corner under compression

The preceding analysis is deliberately abstract: it isolates the structural properties of the energy (Assumptions 1–5) and of the loading (the hypotheses of §4) that the argument actually uses, without committing to a domain. We now fix a concrete material and a concrete family of geometries and ask how much of that hypothesis list can be *discharged* rather than assumed. The material is the compressible neo-Hookean energy (7) with the barrier penalty $f_{\log}$. The geometry is a body whose boundary carries a clamped (Dirichlet) face meeting a traction-free (Neumann) face at a right angle, under a compressive load.

The abstract development of §3–§4 is already dimension-independent: it analyses the transverse problem at the codimension-two Dirichlet–Neumann edge Σ in $\mathbb{R}^d$. Here we instantiate it on concrete bodies and ask how much of that hypothesis list can be *discharged* rather than assumed.

5.1 Geometry and the singular set

We consider two model bodies in $\Omega \subset \mathbb{R}^d$.

- **(H) Half-ball.** $\Omega = \{\, \boldsymbol{x} \in \mathbb{R}^d : |\boldsymbol{x}| < 1,\ x_d > 0 \,\}$. The flat base $\Gamma_D = \{x_d = 0\} \cap \partial\Omega$ is clamped; the spherical cap $\Gamma_N = \{|\boldsymbol{x}| = 1\} \cap \partial\Omega$ is traction-free.
- **(C) Box.** $\Omega = (-1, 1)^{d-1} \times (0, 1)$. The base $\Gamma_D = \{x_d = 0\}$ is clamped; the remaining faces are traction-free.

In both cases the load is a monotone compression transmitted through Γ_N (a pressure or an inward normal displacement on the cap, respectively the lateral/top faces), scaled by the pseudo-time $t \in [0,1]$ of §4.

The *singular set* $\Sigma \subset \partial\Omega$ is the locus where $\overline{\Gamma_D}$ meets $\overline{\Gamma_N}$:

- for (H), $\Sigma = \{|\boldsymbol{x}| = 1,\ x_d = 0\} \cong S^{d-2}$, a *smooth* codimension-two submanifold; the cap is tangent to the vertical at the equator, so it meets the horizontal base at a right angle;
- for (C), Σ is the union of the bottom edges $\{x_d = 0,\ x_i = \pm 1\}$, again right-angle Dirichlet–Neumann edges, *together with* the bottom vertices, where d faces meet.

Fix $\boldsymbol{x}_0 \in \Sigma$ away from any vertex and split $\mathbb{R}^d = T_{\boldsymbol{x}_0}\Sigma \oplus \Pi$, with Π the two-plane normal to Σ. In the slice Π the boundary reduces to a Dirichlet ray and a Neumann ray meeting at $90°$: *exactly* the planar junction of §3.5. We write B_ρ for a tubular neighbourhood of Σ of radius ρ and use (r, θ) for polar coordinates in Π as before.

Remark 11 (Why the half-ball is the clean model). For (H) the singular set Σ is a smooth closed manifold with no vertices, so the singularity is a *pure edge* everywhere on Σ and the transverse problem is the planar right-angle junction at every point. For (C) the edge interiors behave identically, but the bottom vertices carry an additional, generally stronger, polyhedral-vertex singularity [7] that is outside the scope of the edge analysis; one either rounds the vertices or restricts B_ρ to the edge interior. We therefore state the dimension-independent results for (H) and note that they hold verbatim on the edge interiors of (C).

5.2 Tier 1: the material assumptions are already discharged

For the neo-Hookean energy (7) with $f_{\log}$, the standing assumptions of §3 hold with explicit constants and require no information about the domain: Assumption 1 (regularity, objectivity, and the collapse barrier $f_{\log}(J) \to \infty$), Assumption 2 (the inverse-channel bound, with $\Theta = \Theta_{\log}$ of (23)), Assumption 3 and the sharper Lemma 5 (strong ellipticity with constant $\geq \mu_e$ throughout compression), and Assumption 5 (logarithmic growth, $p = 1$) are all established in Lemma 1, Remark 1 and §3.4. None of these proofs uses $d = 2$: the tangent formula (10), the rank-one identity (18), and the growth functions (23) are algebraic identities in $\mathrm{GL}^+(d)$. Thus the entire material side of the hypothesis list is closed for the concrete model, in every dimension. What remains are the hypotheses on the *solution and loading*, which is where the geometry enters.

5.3 The edge exponent and the scenario it selects

By the edge-singularity calculus [6, 7], the leading singular exponents of the linearised elasticity operator along the smooth edge Σ coincide with those of the transverse two-dimensional pencil in Π, modulated smoothly along Σ. That transverse pencil is the planar right-angle Dirichlet–Neumann corner of (13); its leading exponent $\alpha \in (0,1)$ is the smallest root of the corresponding clamped–free Williams secular equation and depends only on the Poisson ratio. Consequently the kinematic blow-up (14), $\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}(r^{\alpha-1})$, holds with a *computable* exponent, and by Theorem 1 the nonlinear tangent inherits exactly this α in the benign regime — on the annulus outside the dilation core of Lemma 3, the core being absent for mild loading or a globally convex-volumetric barrier.

Which scenario of §3 is realized is governed by the sign of the dominant eigenvalue of $\nabla \boldsymbol{u}$ at the edge, equivalently by the sign of the stress-intensity factor K in (13). This is a global quantity, set by the far field, and is not determined by the local pencil alone. The compressive load we impose selects $K < 0$: the material is pushed into the corner, the dominant eigenvalue of $\nabla \boldsymbol{u}$ is negative, and Scenario B (§3.6) is the relevant branch. The quantitative form of "compressive enough" is the hypothesis (SC) below; it simultaneously selects Scenario B and drives the structural instability of §5.4. The complementary tensile load ($K > 0$) selects Scenario A, where (NC) holds and Theorem 1 applies; for the present geometry under compression that case does not arise near the edge.

5.4 Existence of the critical load via a localized surface instability

The one genuinely non-elementary ingredient of §4 was hypothesis (ii) of Lemma 6: the existence of a structurally destabilising field, $Q_{t_*}(\boldsymbol{v}_*) < 0$, which we there declined to establish and flagged as the corner counterpart of the Biot/creasing instability. For the concrete clamped–free geometry we now *prove* it, by reduction to the surface instability of the compressed half-space.

The free face Γ_N under compression is, locally, a traction-free surface compressed parallel to itself — the exact configuration of Biot's classical analysis [1]. A homogeneously compressed neo-Hookean half-space loses stability to a surface mode once the surface-parallel stretch falls below a threshold $\lambda_\star \in (0,1)$ (for the incompressible model $\lambda_\star \approx 0.544$; for the compressible energy (7) the threshold is the corresponding root of the surface secular equation, still in $(0,1)$ [3]). We package the loading requirement as a single hypothesis.

Assumption 8 (Surface compression reaches threshold, (SC))**.** There exist $t_* \in (0,1]$ and an interior point $P \in \Gamma_N \cap B_\rho$, at positive distance from both Σ and $\overline{\Gamma_D}$, such that the fundamental state compresses the free face below the half-space threshold there: the surface-parallel principal stretch of $\boldsymbol{F}_0(t_*)$ at P satisfies $\lambda_\parallel\big(\boldsymbol{F}_0(t_*); P\big) < \lambda_\star$.

(SC) is an input on the *loading*, in the same spirit as the monotonicity condition (M) of Lemma 7: it says only that the prescribed compression is strong enough that the free surface dips below its own buckling threshold somewhere near the edge. Since t merely normalises the load amplitude, (SC) can always be met by loading hard enough, provided the material does not fail elsewhere first.

Lemma 12 (Localized surface instability $\Rightarrow$ structural instability)**.** *Under Assumption 8, hypothesis (ii) of Lemma 6 holds: there is $\boldsymbol{v}_* \in V$, supported in B_ρ and vanishing on Γ_D, with*

$$Q_{t_*}(\boldsymbol{v}_*) = \int_{B_\rho} \nabla \boldsymbol{v}_* : \mathbb{L}\big(\boldsymbol{F}_0(t_*)\big) : \nabla \boldsymbol{v}_* \, dV < 0. \tag{63}$$

Consequently, with $\mu_1(0) > 0$ from Lemma 6(i), there is a first critical load $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$.

Proof. Work in coordinates centred at P: let $n \geq 0$ be the depth into the body along the inward normal to Γ_N and $\boldsymbol{s} \in \mathbb{R}^{d-1}$ the surface-tangent coordinates, so that near P the body is the half-space $\{n > 0\}$ and $\boldsymbol{F}_0(t_*)$ is C^1 with value $\boldsymbol{F}_P := \boldsymbol{F}_0(t_*; P)$ whose surface-parallel stretch is $\lambda_\parallel < \lambda_\star$.

Step 1: the homogeneous unstable mode. Freeze the coefficients at $\boldsymbol{F}_P$ and consider the constant-tensor form $Q_0(\boldsymbol{v}) = \int_{\{n>0\}} \nabla \boldsymbol{v} : \mathbb{L}(\boldsymbol{F}_P) : \nabla \boldsymbol{v} \, dV$ on the half-space, with $\boldsymbol{v}$ traction-free at $n = 0$. Because $\lambda_\parallel < \lambda_\star$, the half-space is past its surface-instability threshold [1, 3]: for each surface wavevector $\boldsymbol{k} \in \mathbb{R}^{d-1}$, $k := |\boldsymbol{k}|$, there is a surface mode

$$\boldsymbol{v}_{\boldsymbol{k}}(\boldsymbol{s}, n) = \mathrm{Re}\left[\, \hat{\boldsymbol{v}}(kn)\, e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{s}} \,\right], \qquad |\hat{\boldsymbol{v}}(\zeta)| \leq C e^{-\beta\zeta} \ (\beta > 0), \tag{64}$$

decaying exponentially into the body on the depth scale $1/k$. Since $\mathbb{L}(\boldsymbol{F}_P)$ is constant, the rescaling $(\boldsymbol{s}, n) \mapsto k^{-1}(\boldsymbol{s}, n)$ shows that over a surface window W of area $|W|$,

$$\int_{\{n>0\}} \mathbf{1}_W \, \nabla \boldsymbol{v}_{\boldsymbol{k}} : \mathbb{L}(\boldsymbol{F}_P) : \nabla \boldsymbol{v}_{\boldsymbol{k}} \, dV = q(\lambda_\parallel) \, k \, |W| \, (1 + o(1)), \qquad q(\lambda_\parallel) < 0, \tag{65}$$

where q is the Biot secular form, strictly negative past threshold. Likewise $\int_{\{n>0\}} \mathbf{1}_W \, |\nabla \boldsymbol{v}_{\boldsymbol{k}}|^2 \sim k \, |W|$ and $\int_{\{n>0\}} \mathbf{1}_W \, |\boldsymbol{v}_{\boldsymbol{k}}|^2 \sim k^{-1}|W|$, the latter two by the same scaling.

Step 2: cut-off. Let χ be smooth, $\chi \equiv 1$ on $B(P, \delta/2)$, $\operatorname{supp} \chi \subset B(P, \delta)$, $|\nabla \chi| \leq C/\delta$, with δ small enough that $B(P, \delta) \cap \overline{\Omega}$ avoids both Σ and $\overline{\Gamma_D}$ and lies in B_ρ (possible since P is at positive distance from both). Set $\boldsymbol{v}_* := \chi \, \boldsymbol{v}_{\boldsymbol{k}}$. Then $\boldsymbol{v}_* \in V$ and vanishes on Γ_D (its support avoids $\overline{\Gamma_D}$); it is unconstrained on the free face, as required.

Step 3: error control. Write $\nabla \boldsymbol{v}_* = \chi \nabla \boldsymbol{v}_{\boldsymbol{k}} + (\nabla \chi) \otimes \boldsymbol{v}_{\boldsymbol{k}}$ and expand (63). The effective window is $W \sim B(P, \delta) \cap \Gamma_N$, of area $|W| \sim \delta^{d-1}$, and the decay confines the mass to depth $1/k$. Three errors arise.

(a) *Coefficient variation:* replacing $\mathbb{L}(\boldsymbol{F}_0(\boldsymbol{x}))$ by $\mathbb{L}(\boldsymbol{F}_P)$ costs $\leq \sup_{B(P,\delta)} \|\mathbb{L}(\boldsymbol{F}_0) - \mathbb{L}(\boldsymbol{F}_P)\| \int \chi^2 |\nabla \boldsymbol{v}_{\boldsymbol{k}}|^2 \leq C\delta \cdot k\delta^{d-1} = Ck\delta^d$ (by C^1 regularity of $\boldsymbol{F}_0$ at P).

(b) *Gradient commutator:* the cross term and $|\nabla \chi|^2 |\boldsymbol{v}_{\boldsymbol{k}}|^2$ term are bounded by $C\big(\delta^{-1} \int_W |\boldsymbol{v}_{\boldsymbol{k}}| |\nabla \boldsymbol{v}_{\boldsymbol{k}}| + \delta^{-2} \int_W |\boldsymbol{v}_{\boldsymbol{k}}|^2\big) \leq C\big(\delta^{d-2} + \delta^{d-3} k^{-1}\big)$.

(c) *Window vs. cut-off:* replacing $\mathbf{1}_W$ by χ^2 in (65) changes the leading constant by $1 + o(1)$ only.

The leading term is $q(\lambda_\parallel) \, k \, \delta^{d-1} < 0$. The errors (a)–(b), relative to it, are $\mathcal{O}(\delta)$, $\mathcal{O}((k\delta)^{-1})$ and $\mathcal{O}((k\delta)^{-2})$. Choosing $\delta = k^{-1/2}$ sends $\delta \to 0$ and $k\delta = k^{1/2} \to \infty$, so all relative errors vanish as $k \to \infty$. Hence for k large enough,

$$Q_{t_*}(\boldsymbol{v}_*) = q(\lambda_\parallel) \, k \, \delta^{d-1} \, (1 + o(1)) < 0,$$

which is (63). The final statement is Lemma 6 with hypothesis (ii) now supplied. □ □

Remark 12 (What this closes). Lemma 12 converts the existence of the critical load t_c — previously conditional on the unproved hypothesis (ii) — into a *theorem* for the concrete clamped–free geometry, contingent only on the loading hypothesis (SC). The mechanism is made explicit: the corner instability *is* the half-space surface (creasing) instability, localised to the free face near the edge by the cut-off (64)–(65). This is the precise sense in which the corner bifurcation is “the corner analogue of creasing.”

Remark 13 (On the threshold $\lambda_\star$: compressible vs. incompressible). The proof uses only the *existence* of a surface-instability threshold $\lambda_\star \in (0, 1)$, not its value. The familiar number $\lambda_\star \approx 0.544$ is Biot's result for the *incompressible* neo-Hookean half-space [1]; the present material (7) is compressible, and its threshold is the corresponding root of the compressible surface secular equation, which depends on the ratio λ_e/μ_e (equivalently the Poisson ratio) and tends to 0.544 in the incompressible limit $\lambda_e/\mu_e \to \infty$ [3]. This is the same instability with a material-dependent threshold, not a discrepancy: Lemma 12 *reduces* the corner instability to the half-space one and thereby agrees with the creasing literature rather than competing with it. Accordingly we keep $\lambda_\star$ symbolic throughout.

5.5 Symmetry: simplicity of the mode and the vanishing quadratic

Both model bodies of §5.1 possess reflection symmetries that the fundamental compressive state inherits: (H) is invariant under every orthogonal map fixing the x_d-axis, and (C) under each coordinate reflection $x_i \mapsto -x_i$ ($i < d$). Let R be such a reflection and $\mathcal{R}$ its action on displacement fields. Because $\boldsymbol{F}_0(t)$ and hence $\mathbb{L}(\boldsymbol{F}_0(t))$ are R-invariant, the second-variation operator $\mathcal{L}(t)$ commutes with $\mathcal{R}$, so its eigenspaces split into the symmetric ($\mathcal{R} = +1$) and antisymmetric ($\mathcal{R} = -1$) sectors. This yields, for the concrete geometry, two of the ingredients that §4 had to posit:

- *Simplicity (part of Assumption 6).* The lowest eigenvalue within a fixed symmetry sector is generically simple, and the critical mode $\boldsymbol{\phi}$ is the ground state of one sector; the genericity is the standard one for self-adjoint operators commuting with a finite symmetry group. Thus simplicity need not be assumed separately — only the spectral *attainment* (the borderline-elliptic point of Remark 6) remains genuinely open.

- *Vanishing quadratic coefficient (part of Assumption 7).* The reduced potential of Proposition 1 inherits the $\mathcal{R}$-symmetry of the perfect state. If $\boldsymbol{\phi}$ lies in the antisymmetric sector, $w \mapsto -w$ is a symmetry of the reduced equation and the quadratic coefficient vanishes. For the neo-Hookean model this is made fully explicit in Lemma 13: c_2 is odd in the first-order volume change D_1, which R flips, so $c_2 = 0$ exactly. The relevant reflection R is the lateral, *face-preserving* symmetry of the model bodies of §5.1, *not* the corner bisector (which would interchange Γ_D and Γ_N and is no symmetry of the mixed problem).

Lemma 13 (Neo-Hookean reduced coefficients are volumetric; the quadratic vanishes)**.** *For the neo-Hookean energy* (7) *the deviatoric term* $\frac{\mu_e}{2}(\operatorname{tr} \boldsymbol{F}^T\boldsymbol{F} - d)$ *is quadratic in* $\boldsymbol{F}$*, so its third and higher Fréchet derivatives vanish; hence* every *reduced bifurcation coefficient of order* ≥ 3 *is purely volumetric, determined by* f *alone. Writing* $J(w) = \det(\boldsymbol{F}_0 + w\nabla\boldsymbol{\phi}) = \sum_k D_k\, w^k$ *with* D_k *the mixed discriminants of Lemma 9 (*$D_1 = \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi}$*,* $D_2 = d_\phi$*,* $D_k = 0$ *for* $k > d$*) and* $f^{(k)} = f^{(k)}(J_0)$*, the single-mode coefficients are*

$$c_2 = \tfrac{1}{2}\int_{B_\rho} \left[f''' D_1^3 + 6f'' D_1 D_2 + 6f' D_3\right] dV, \tag{66}$$

$$b_{\mathrm{dir}} = \tfrac{1}{6}\int_{B_\rho} \left[f'''' D_1^4 + 12 f''' D_1^2 D_2 + 12 f'' D_2^2 + 24 f'' D_1 D_3 + 24 f' D_4\right] dV. \tag{67}$$

Two consequences hold for the concrete geometry, whose fundamental state is R-symmetric.

(i) The quadratic vanishes, exactly and without correction. *As a* third-*order quantity* c_2 *receives no orthogonal-mode (Lyapunov–Schmidt) feedback — that correction first enters at fourth order — so* (66) *is the full* c_2*. Every term of* (66) *is* odd *in* D_1*, and* D_1 *is* R*-odd for an* R*-antisymmetric mode while* $f^{(k)}(J_0)$ *and the even discriminants* D_2 *are* R*-even; the integrand is therefore odd over the* R*-symmetric* B_ρ *and* $c_2 = 0$*. This* proves *the vanishing-quadratic clause of Assumption 7.*

(ii) The frozen cubic and quintic are sign-definite, and adverse to subcriticality. *With* D_1 *asymptotically small (Lemma 10) and* $d_\phi > 0$ *(Lemma 11), the leading frozen parts are* $b_{\mathrm{dir}} = 2\int_{B_\rho} f''(J_0)\, d_\phi^2\, dV$ *and* $d_{\mathrm{dir}} = \int_{B_\rho} f'''(J_0)\, d_\phi^3\, dV$*. For the logarithmic penalty* $f''_{\log}(J_0) = (\mu_e + 2\lambda_e - 2\lambda_e \log J_0)/J_0^2 > 0$ *and* $f'''_{\log}(J_0) = (4\lambda_e \log J_0 - 6\lambda_e - 2\mu_e)/J_0^3 < 0$ *throughout compression* $J_0 < 1$*, so* $b_{\mathrm{dir}} > 0$ *and* $d_{\mathrm{dir}} < 0$ *— the* opposite *of the subcritical pattern* $b < 0 < d$*.*

Proof. Expand $\Psi(\boldsymbol{F}_0+w\nabla\boldsymbol{\phi}) = \frac{\mu_e}{2}(\operatorname{tr}\boldsymbol{F}^T\boldsymbol{F}-d)+f(J(w))$ in w. The deviatoric part is a polynomial of degree two in w, so it contributes only to the coefficients of w^0, w^1, w^2; all coefficients of order ≥ 3 come from $f(J(w))$, whose w-derivatives are given by Faà di Bruno's formula in the J-derivatives $f^{(k)}$ and the discriminants $D_k = \frac{1}{k!}\partial_w^k \det(\boldsymbol{F}_0+w\nabla\boldsymbol{\phi})|_0$. Reading off the third and fourth derivatives gives (66)–(67); statements (i)–(ii) are then as stated, using $f''_{\log} > 0$, $f'''_{\log} < 0$ on $J_0 < 1$. □ □

What symmetry fixes is thus the quadratic coefficient; what it does *not* fix is the sign of the cubic b and quintic d. Indeed the explicit single-mode parts (67) point *against* subcriticality ($b_{\text{dir}} > 0$, $d_{\text{dir}} < 0$ by Lemma 13(ii)), so the subcritical pattern $b < 0 < d$ rests entirely on the geometry-dependent orthogonal-mode feedback — the irreducible core, which we settle numerically rather than by analogy (Remark 8).

5.6 Kinematic relief

The kinematic relief is already dimension-independent, and — in contrast to the original formulation — it no longer rests on an assumed rotation condition. Earlier drafts *posited* that the critical mode be exactly volume-preserving at first order, $\operatorname{cof}\boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = 0$. That hypothesis is now *discharged*: Lemma 10 bounds the first-order volume change by the barrier alone, $D_1 = \operatorname{cof}\boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ — small with J_0, but not assumed zero — and Lemma 9 then proves the unconditional lower bound

$$J = J_0 + w\,D_1 + w^2 d_\phi + \mathcal{O}(w^3) \;\geq\; \tfrac{1}{2}J_0 + \tfrac{1}{2}w^2 d_\phi \qquad \text{in } \mathbb{R}^d, \tag{68}$$

with d_ϕ the second mixed discriminant of $\boldsymbol{F}_0$ and $\nabla\boldsymbol{\phi}$ (the second-order coefficient of the determinant, $= \det\nabla\boldsymbol{\phi}$ for $d = 2$). The proof of Lemma 10 uses only $f''(J)(\delta J)^2 \leq C$ and is valid for all d, so the relief is genuinely a theorem in every dimension rather than a kinematic assumption. For the concrete model the only points to add are that the genericity $d_\phi > 0$ holds off a measure-zero set of modes, by the transversality argument of Lemma 11, and that the collapse locus Σ_{c} is generically the codimension-one set inside B_ρ along which the relieved branch would still collapse (or, when the severest compression sits at the vertex, the point $\Sigma_{\text{c}} = \{0\}$ of codimension two, Remark 10, which only sharpens the integrability); the regularised bound of Theorem 2 and the localisation of Proposition 4 then apply verbatim.

5.7 Attainment of the critical mode

The second half of Assumption 6 — that $\mu_1(t_c) = 0$ is *attained* by a genuine eigenmode rather than being the bottom of essential spectrum — looks like the hard, borderline-elliptic point of Remark 6. It is in fact a consequence of a single, transparent geometric condition: that the *buckling precedes the pointwise collapse* of the fundamental state. The essential-spectrum threat is tied to the scale-invariance of the corner at collapse (the same wavenumber degeneracy that makes flat creasing scale-free, and that let $k \to \infty$ in Lemma 12); it is activated only when $\mathbb{L}$ blows up, i.e. only when $\lambda_{\min} \to 0$. So long as the crossing occurs while the fundamental stretch is still bounded below, the operator has bounded coefficients and the spectrum is discrete.

Assumption 9 (Buckling precedes collapse, (BC))**.** The fundamental path keeps the stretch off collapse up to the crossing:

$$\inf_{t\leq t_c}\ \inf_{\boldsymbol{x}\in\overline{B_\rho}}\ \lambda_{\min}\big(\boldsymbol{F}_0(t,\boldsymbol{x})\big) \;\geq\; c > 0. \tag{69}$$

Lemma 14 (Attainment under (BC))**.** *Under Assumption 9, for every $t \le t_c$ the tangent $\mathbb{L}(\boldsymbol{F}_0(t))$ is bounded on B_ρ, the operator $\mathcal{L}_c = \partial_{\boldsymbol{u}} G(\boldsymbol{u}_0(t_c), t_c, 0)$ has compact resolvent and discrete spectrum, and the critical value $\mu_1(t_c) = 0$ is attained by an eigenmode $\boldsymbol{\phi} \in V$. Together with the simplicity argument of §5.5, this discharges Assumption 6 in full.*

Proof. By (69), $\|\boldsymbol{F}_0(t)^{-1}\| = \lambda_{\min}^{-1} \le c^{-1}$ and $\det \boldsymbol{F}_0(t)$ is bounded on the fundamental path; hence by the structural bound (11) with $\Theta = \Theta_{\log}$, $\|\mathbb{L}(\boldsymbol{F}_0(t))\| \le \mu_e + c^{-2}\Theta_{\log}(\det \boldsymbol{F}_0)$ is uniformly bounded on B_ρ for $t \le t_c$. By Lemma 5 the same tensor is strongly elliptic with constant $\ge \mu_e$ throughout compression; with Korn's inequality the second-variation form obeys a Gårding estimate $Q_{t_c}(\boldsymbol{v}) \ge \gamma \|\boldsymbol{v}\|^2_{H^1(B_\rho)} - C\|\boldsymbol{v}\|^2_{L^2(B_\rho)}$ on $V \subset H^1$. Thus $\mathcal{L}_c + C$ is coercive on H^1, its inverse maps L^2 boundedly into H^1, and the Rellich embedding $H^1(B_\rho) \hookrightarrow\hookrightarrow L^2(B_\rho)$ — valid on the bounded domain, the edge Σ notwithstanding — makes the resolvent of $\mathcal{L}_c$ compact. A self-adjoint operator with compact resolvent has purely discrete spectrum and attains its lowest eigenvalue; at t_c that eigenvalue is $\mu_1(t_c) = 0$, with eigenmode $\boldsymbol{\phi}$. □ □

Remark 14 (What (BC) costs, and why it is the right residue)*.* Lemma 14 relocates the difficulty from a functional-analytic statement about essential spectrum to a comparison of two load thresholds: the buckling load t_c and the pointwise-collapse load $t_{\text{coll}} := \inf\{t : \inf_{B_\rho} \lambda_{\min}(\boldsymbol{F}_0(t)) = 0\}$, with (BC) asking $t_c < t_{\text{coll}}$. This is genuinely cleaner than "attainment": (i) Lemma 12 already bounds t_c from above by the surface-creasing load t_*, attained while $\lambda_{\min} \approx \lambda_\star = \mathcal{O}(1)$ at the creasing point — not at collapse; (ii) the barrier $f_{\log}(J) \to \infty$ makes $J = 0$ cost infinite energy, so the true (not linearly-predicted) fundamental path resists pointwise collapse and pushes t_{coll} up; and (iii) (BC) is directly checkable in a computation, by monitoring $\min_{\boldsymbol{x}} \lambda_{\min}$ at the detected crossing. The one place it can fail is the edge tip, where the *linear* prediction $\lambda_{\min} \sim 1 - |K| r^{\alpha-1}$ wants to collapse for any load; bounding the barrier-regularised tip collapse load from below is the concrete content left open, in place of the opaque spectral hypothesis.

5.8 The irreducible core

Table 1 summarises the effect of fixing the concrete model. Of the hypotheses of §3–§4, the material assumptions are discharged by the neo-Hookean structure (§5.2), the existence of the critical load by the localized-creasing Lemma 12 under the loading hypothesis (SC), mode simplicity and the vanishing quadratic by reflection symmetry (§5.5), and mode attainment by Lemma 14 under the geometric condition (BC). What remains are two hypotheses, both now in concrete, checkable form rather than as opaque spectral postulates:

1. *Buckling precedes collapse*, (BC) (Assumption 9): the threshold inequality $t_c < t_{\text{coll}}$ that powers attainment (Lemma 14). It is bounded on one side by Lemma 12 ($t_c \le t_*$, at $\lambda_{\min} = \mathcal{O}(1)$) and supported by the energy barrier (Remark 14); the residual content is a lower bound on the barrier-regularised tip collapse load, which we verify numerically.

2. *Subcritical sign pattern*, $b < 0 < d$ (Assumption 7): the directions in which the cubic and quintic bend the branch. Here the neo-Hookean structure is doubly informative. Lemma 13 reduces the coefficients to volumetric integrals and shows the explicit single-mode parts are *adverse* to subcriticality ($b_{\text{dir}} > 0$, $d_{\text{dir}} < 0$ in compression), so $b < 0 < d$ can come only from the geometry-dependent orthogonal-mode feedback. The creasing literature [1, 3] establishes subcriticality for the flat surface, but the sign at a *corner* is delivered neither by symmetry nor by the frozen terms; we evaluate the feedback-corrected coefficients numerically (Remark 8).

Hypothesis	Abstract status	Status for the concrete model
Reg./growth/ellip./mild (Ass. 1–5)	standing assumptions	**proved** (Lem. 1, §5.2)
Edge exponent α, scenario selection	abstract pencil	**computed**; B selected by $K < 0$
Critical load t_c exists (Lem. 6(ii))	not established	**proved** under (SC) (Lem. 12)
Transversality (M) (Lem. 7)	input on loading	**holds** for monotone compression
Mode simplicity (Ass. 6)	assumed	**generic** per symmetry sector (§5.
Quadratic coeff. vanishes (Ass. 7)	assumed by symmetry	**proved**: c_2 odd in D_1, R-forced (
Mode attainment (Ass. 6)	assumed	**proved** under (BC) (Lem. 14)
First-order relief / rotation (Lem. 10, 9)	was posited; now **proved** (all d)	genericity $d_\phi > 0$ (§5.6)
Buckling precedes collapse (BC) (Ass. 9)	—	*numerical*: $t_c < t_{\mathrm{coll}}$ (Rem. 14)
Subcriticality $b < 0 < d$ (Ass. 7)	assumed	*numerical*; frozen parts adverse (I

Table 1: Hypothesis ledger for the concrete clamped–free neo-Hookean model. Fixing the material and geometry discharges every spectral postulate; the residue is two concrete, numerically checkable conditions — that buckling precedes pointwise collapse, and that the reduced cubic is subcritical.

The upshot is that for the clamped–free neo-Hookean corner under compression the abstract picture of §3–§4 is, but for two clearly delimited conditions, a sequence of theorems: the material bounds are explicit, the corner is genuinely compressive, the instability is a localized creasing event with a constructed destabilising mode, the critical mode is simple and attained (no quadratic term, by symmetry), and the kinematic relief that regularises the tangent holds in every dimension. Notably, the spectral postulates of the abstract theory — attainment and simplicity — are no longer assumed but proved, and the first-order volume preservation (rotation condition) that the relief once posited is now itself a barrier-forced theorem (§5.6); what remains are two *concrete* and numerically checkable conditions: that buckling precedes pointwise collapse ($t_c < t_{\mathrm{coll}}$), and that the reduced cubic is subcritical ($b < 0$). Both are exactly what the creasing problem on a flat surface also leaves to computation, and they are inherited here rather than created by the corner.

6 Conclusions

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